Six-Max Preflop Equilibria via Monte Carlo CFR

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Abstract

We study the preflop stage of six-player no-limit Texas hold'em cash games at 100 big blinds as a finite game in its own right, and compute approximate equilibria for it. The game has one raise size per level, cold calls and squeezes, and rake; its 7,499 decision nodes end either in a fold, a called all-in, or the flop. At the flop we stop and pay each player a share of the pot given by their equity scaled by a *realization factor*, a parametric model of how well a hand converts equity into winnings after the flop, built from suitedness, connectivity (including wheel straights), pairs, position, initiative, domination and pot depth.

We solve the game with external-sampling Monte Carlo counterfactual regret minimisation with linear averaging, which solves the full tree in about a minute on a laptop. On a heads-up push/fold game, where convergence is guaranteed, exploitability falls to 1.5×10^{-5} big blinds per hand. The realization parameters are fitted to published equilibrium charts by coordinate search over full solves; the solution's category-balanced distance to the charts is 0.136, against 0.319 for a position-only model. A solve with the charts locked shows that the remaining disagreement is concentrated in how often offsuit aces and broadways re-raise. We also study a cheaper fit that holds strategies fixed and treats the charts as quantal responses: it improves the frequent spots but cannot see equilibrium feedback in 4-bet pots, and moving halfway towards it gives a small gain of about two standard errors. An interactive trainer grades decisions against the solution with importance-sampled rollouts.

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Notation

Symbol	Meaning
$N = \{1, \dots, n\}$, c	players; the chance player
H, Z	decision histories; terminal histories
$P(h)$, $A(h)$	player to act at h; legal actions at h
$\mathcal{I}_i$, I	player i's information partition; an information set
σ, σ^t, $\bar{\sigma}$	a behavioural strategy profile; the iterate at time t; the average strategy
$\pi^\sigma(h)$, $\pi_i^\sigma(h)$, $\pi_{-i}^\sigma(h)$	reach probability; player i's part; everyone else's (chance included)
u_i	utility of player i
$v_i^\sigma(I)$, $v_i^\sigma(I, a)$	counterfactual value of I; with a forced at I
$R^T(I, a)$, R_+^T	cumulative counterfactual regret; its positive part
$\varepsilon(\sigma)$	exploitability
$k \in \{0, \dots, 168\}$, $b(x)$, n_k	hand class; the class of combo x; combos in class k (6, 4 or 12)
eq_i	equity of player i
pot, C_i, rake(z)	pot; player i's contribution; rake taken at terminal z
ρ_i, θ	realization factor of player i; the realization parameters
SPR, γ, κ	stack-to-pot ratio at the flop; depth fade $\gamma = \mathrm{SPR}/(\mathrm{SPR} + \kappa)$; its half-life
$\phi, \eta, \varsigma, \delta, \delta_{\mathrm{rr}}, \lambda, \beta, \iota, \psi, \xi, \chi, \alpha_{\mathrm{oop}}$	realization parameters (Table 3.1)
s_i	realization-adjusted pot share
w_t, $S(I)$	weight of iteration t (its block number); strategy sum at I
ℓ, N_{eff}	likelihood weight of a deal; Kish's effective sample size
D_s, $J(\theta)$	weighted total-variation distance at spot s; tuning objective
$q_\theta(a \mid s, k)$, $\mathcal{L}(\theta)$	logit (quantal) response; fixed-strategy loss
τ	softmax temperature
ϵ	tolerance of an ϵ-equilibrium (distinct from the exploitability ε)

Chapter 1

Introduction

1.1 Motivation and scope

Poker has long served as a benchmark for reasoning under imperfect information [1]. Unlike chess or Go, a player does not see the whole state of the game: the opponents' cards are hidden, so good play has to be randomised, and the value of an action depends on what the opponents believe as much as on the cards themselves. The last decade has seen a series of landmark results built on one family of algorithms, *counterfactual regret minimisation* (CFR) [35]. Heads-up limit hold'em was essentially solved by Cepheus [3]. DeepStack [24] and Libratus [5] beat professional players at heads-up no-limit hold'em, and Pluribus [7] did so in the six-player game.

These systems need large compute budgets and solve the game street by street, re-solving subgames as play unfolds. Human players study a far smaller object: *preflop charts*, tables of what to do with each starting hand before any community cards are dealt. Preflop is the decision every player faces in every hand, the stage where most hands end, and the one whose structure is simple enough to tabulate. Commercial tools publish such charts, computed by solving the full game on large clusters, but the charts come without the model behind them.

This report studies the preflop stage of six-player ('six-max') no-limit hold'em cash games at a stack depth of 100 big blinds as a finite game in its own right. We

(i) define a preflop-only game whose payoffs at the moment the flop would be dealt approximate the value of playing the rest of the hand, through a parametric *realization model*;

(ii) solve it with external-sampling Monte Carlo CFR [20] with linear averaging [6], on a laptop, in about a minute;

(iii) fit the realization model's parameters so that the solution matches published equilibrium charts [29], and study a cheaper fitting method that avoids a solve per candidate;

(iv) build an interactive trainer on top of the solution that grades a user's decisions by importance-sampled rollouts.

Everything is implemented in our own software, Preflop, written in C++20 with a raylib front end [31] and the PokerHandEvaluator library [21], with Python tooling for the fitting.

The equilibrium we compute is an equilibrium of *our* model, not of real poker. How close it gets to charts computed from the full game is therefore the central empirical question, and the realization model is where most of the modelling effort goes.

1.2 A short introduction to six-max hold'em

We assume no knowledge of poker; Sklansky [32] and Chen and Ankenman [10] are standard references for the game and its mathematics.

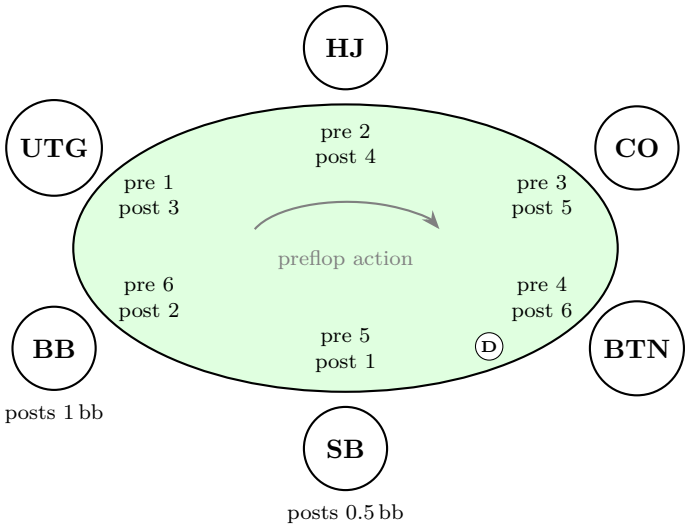


Figure 1.1: A six-max table. The numbers give each seat's order of action preflop ('pre') and on every later street ('post'). D marks the dealer button. Action moves clockwise.

Cards and hands. Each player is dealt two private *hole cards* from a standard 52-card deck. Up to five *community cards* are then dealt face up in three stages: the *flop* (three cards), the *turn* and the *river* (one card each). A player's hand is the best five-card combination of their two hole cards and the five community cards. Hands are ranked from high card, one pair and two pair, through three of a kind (a *set* when it uses a pocket pair), a straight (five consecutive ranks, the ace playing high or low, so A-2-3-4-5, the *wheel*, counts) and a flush (five cards of one suit), up to a full house, four of a kind and a straight flush.

Betting. There are four betting rounds: one before the flop (*preflop*) and one after each of the flop, turn and river. In each round a player facing a bet may *fold* (give up the hand and anything already put in), *call* (match the bet) or *raise* (increase it). In no-limit hold'em a raise may be any size up to the player's whole stack. A hand ends when all but one player have folded, who wins the pot uncontested, or at a showdown after the river, where the best hand wins.

Positions and blinds. At a six-player table the seats are named by their position relative to the *button* (Figure 1.1): under the gun (UTG, also called the lojack, LJ), hijack (HJ), cutoff (CO), button (BTN), small blind (SB) and big blind (BB). Before the cards are dealt, SB and BB post forced bets of 0.5 and 1 big blind (bb), which give the others something to win. Preflop, action starts with UTG and runs clockwise, so the blinds act last. On every later street the order changes: SB acts first and BTN last. Acting last postflop, being *in position* (IP) against the remaining players, is a large advantage, since that player sees what everyone else did before deciding. All amounts in this report are in big blinds, and all stacks start at 100 bb.

Preflop vocabulary. The first voluntary raise is an *open*; folding to it or calling it are the alternatives for the players behind. A re-raise of an open is a *3-bet*, a re-raise of that a *4-bet*, and so on (the blinds count as the first bet). Calling an open without having put money in yet is a *cold call* or *flat*; a 3-bet after an open and one or more calls is a *squeeze*. A 5-bet is all-in. A called 4-bet does not commit the whole stack, but it leaves only about 1.5 to 2 times the pot behind at the flop, so little room for postflop play. A *spot* is a decision point identified by the action before it, such as 'BB facing a BTN open'. A player's *range* at a spot is the set of hands they arrive there with, each weighted by how often. A re-raising range is *polarised* when it holds the strongest hands together with some weaker ones, the middle calling or folding, and *linear* when it is simply the top of the range.

Starting hands. There are $\binom{52}{2} = 1326$ two-card *combos*. Before the flop the suits are interchangeable, so combos fall into 169 *hand classes*: 13 pocket pairs (6 combos each, e.g. AA), 78 suited non-pairs (4 combos, e.g. AKs) and 78 offsuit non-pairs (12 combos, e.g. AKo). We draw them on the usual 13×13 grid: pairs on the diagonal, suited hands above it, offsuit hands below. The ranks ten to ace are the *broadway* cards, and a hand of two of them, such as KQo, is a broadway. In a non-pair the lower card is the *kicker*: AJ and AT both make top pair on an ace-high flop, but AJ has the better kicker and wins when they meet.

Equity and realization. A hand's *equity* against given opponent hands is its expected share of the pot if all remaining community cards were dealt and the hand went to showdown with no further betting. Equity is easy to compute but overstates what a hand actually wins: postflop betting forces folds, and a hand out of position, or one that rarely makes strong combinations, wins less than its equity. Players speak of how well a hand *realises* its equity. Suited and connected hands make flushes and straights and realise well, offsuit hands with a weak kicker realise badly, and position helps everyone. This is the idea our payoff model formalises (Sections 2.6.2 and 3.3).

1.3 The game we solve

Players and stakes. Six players with 100 bb each, blinds 0.5/1, stacks reset every hand. The game ends at the flop: whenever betting closes with two or more players left, we stop and pay out according to a model of the remaining play (Section 2.6.2).

Actions and sizes. Every decision is a subset of $\{F, C, R\}$ (fold, call, raise) with one raise size per level, chosen to match the reference charts [29]:

- opens are to 2.5 bb, or 3 bb from SB;
- 3-bets are 3× the open in position and 5× from the blinds, plus one open size per caller (a squeeze);
- 4-bets are 2× the 3-bet in position and 2.55× out of position;
- a 5-bet is all-in, and facing an all-in a player may only call or fold.

Raise totals are rounded to 0.1 bb. For example, BB 3-bets a BTN open to 12.5 bb, and BTN can 4-bet to 25 bb. Appendix A lists every size.

Cold calling. Players may call an open, and later players may overcall it, but once a 3-bet is in, players who have not yet put money in may only fold or raise. A player who cold-called the open may also call a squeeze. Limping (calling the big blind without raising) is not allowed.

Rake. As in the online games the reference charts are computed for, the house takes 5% of the pot, capped at 4 bb, but only when a flop is dealt or an all-in is called ('no flop, no drop'). The game is therefore negative-sum for the players.

Size. With these rules the betting tree has 7,499 decision nodes and 8,529 terminal histories: 1,703 where everyone else folds, 5,797 called all-ins and 1,029 that reach a flop, 378 of them with three or more players. The longest line has 19 actions. A fragment of the tree is shown in Figure 1.2. With one information set per decision node and hand class, the game has $7{,}499 \times 169 \approx 1.27 \times 10^6$ information sets in total across all seats.

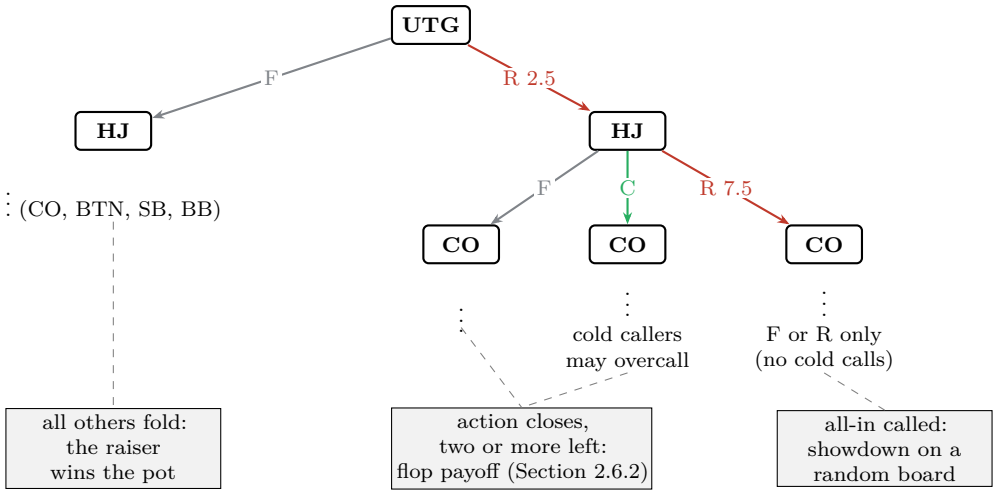


Figure 1.2: A fragment of the preflop betting tree from UTG's first decision. Each decision node is one history; every node is paired with the actor's hand class to give an information set. The three kinds of terminal are shown at the bottom.

1.4 Abstractions

Real no-limit hold'em is far too large to solve directly: even heads-up it has more than 10^{160} decision points [17]. Our game differs from the real game in three ways.

Action abstraction. One raise size per level, no limping, and no cold calls of 3-bets. These match the reference charts' own rules, so they are restrictions shared with the target rather than errors relative to it. Abstraction can have counter-intuitive effects on the quality of an equilibrium [34], which is one reason we keep the rules identical to the charts'.

Card abstraction. Strategies depend on the hand class, not the combo. Preflop, this loses nothing: relabelling suits maps the game onto itself, so an equilibrium can be taken suit-symmetric [14]. The solver still deals real combos from one deck, so *card removal* (holding an ace makes the opponents' aces less likely) is modelled exactly.

Payoff abstraction. This is the large one. We do not play the flop, turn and river. Instead, a terminal where the flop would be dealt pays each live player a share of the pot given by their equity, scaled by a *realization factor* that depends on the hand, the player's role (position, caller or raiser) and the depth of the pot. The factors have a dozen or so parameters, which we fit to the charts (Sections 3.3 and 3.7).

1.5 Related work

CFR was introduced by Zinkevich et al. [35], and its Monte Carlo variants by Lanctot et al. [20]. CFR$^+$ [33] solved heads-up limit hold'em [3], and linear and discounted weighting schemes [6] speed up convergence further. Neller and Lanctot [26] give an accessible tutorial. For more than two players CFR has no equilibrium guarantee, but it still removes dominated actions [13] and produces strong play in practice: Pluribus [7] used Monte Carlo CFR with linear weighting for its six-player blueprint. Function approximation has also been combined with CFR [4, 24].

Preflop-only games have been studied in the form of *push/fold* (all-in or fold) games for short stacks, where the equilibrium can be computed exactly for two players and approximately for three [12]. Our game is much richer: deep stacks, four raise levels, cold calls and squeezes, with postflop value replaced by a model. Fitting payoff parameters so that observed play becomes

rational is a case of inverse game theory, related to learning game parameters from equilibrium play [22] and to quantal-response models of behaviour [23].

1.6 Contributions and outline

The contributions of this report are:

- a precise definition of a six-player preflop game with rake and a realization-adjusted payoff model built from hand features (suitedness, connectivity including wheel straights, pairs, position, initiative, domination and pot depth);
- a fast external-sampling MCCFR solver with lazy linear weighting and lock-free threading, checked against an exactly solvable heads-up game;
- a fit of the model to published charts, a solve with the charts locked that exposes where model and charts disagree, and a study of a cheap fixed-strategy fitting method, including a conjectured reason why it fails where it does;
- an interactive trainer that grades decisions with importance-sampled rollouts and standard errors.

Chapter 2 develops the mathematics from first principles: extensive-form games and equilibria, regret, CFR and its Monte Carlo variants, the estimators we use, and the statistics of fitting. Chapter 3 describes the model and its implementation: the encoding of the game, equity, the realization features, the solver, the reference charts, both fitting methods and the trainer. Chapter 4 reports correctness checks, convergence, agreement with the charts, where the model and the charts disagree, and the outcome of the fitting study. The appendices list the bet sizes, the fitted parameters and further range charts.

Chapter 2

Mathematical Background

This chapter covers games and equilibria (Section 2.1), counterfactual regret minimisation (CFR) and the variants we use (Sections 2.2 to 2.4), what survives with more than two players (Section 2.5), the estimators inside our payoffs and rollouts (Section 2.6), and the fitting of the payoff model (Section 2.7). The poker-specific model is left to Chapter 3: here it is enough that the game is finite and its payoffs computable.

2.1 Preliminaries

2.1.1 Extensive-form games

Definition 2.1 (Finite extensive-form game)**.** A finite extensive-form game with chance and imperfect information consists of

- a finite set of players $N = \{1, \dots, n\}$ and a chance player c;
- a finite set of *histories*, sequences of actions closed under taking prefixes, with $h \sqsubseteq h'$ meaning h is a prefix of h'. The histories with no extension are the *terminal* histories Z; the others are the decision histories H, and $A(h) = \{a : ha \text{ is a history}\}$ are the legal actions at h;
- a player function $P : H \to N \cup \{c\}$, and for every h with $P(h) = c$ a probability distribution $\sigma_c(h, \cdot)$ on $A(h)$;
- utilities $u_i : Z \to \mathbb{R}$ for $i \in N$;
- for each player i an *information partition* $\mathcal{I}_i$ of $\{h \in H : P(h) = i\}$, such that $A(h) = A(h')$ whenever h and h' lie in the same *information set* $I \in \mathcal{I}_i$. We write $A(I)$ for this common set.

An information set collects the histories that player i cannot tell apart when it is their turn: in poker, those that differ only in the other players' cards. Player i has *perfect recall* if any two histories in the same $I \in \mathcal{I}_i$ pass through the same sequence of i's own information sets and actions: a player never forgets what they knew or did. All games in this report have perfect recall. In ours an information set is a pair of a public betting history and the acting player's hand class (Section 3.1), and the class does not change during the hand. The game is *zero-sum* if $\sum_i u_i(z) = 0$ for every $z \in Z$, and *two-player zero-sum* (2p0s) if moreover $n = 2$.

Example 2.2 (A bluffing game)**.** Both players ante one chip. Chance deals player 1 a High or a Low card with probability $\frac{1}{2}$ each. Player 1 checks, ending the hand at a showdown worth ± 1, or bets one chip. Facing a bet, player 2, who has not seen the card, folds (player 1 wins the ante, $+1$) or calls (showdown for ± 2). Figure 2.1 shows the tree. Player 1 has two singleton information sets, one per card; player 2 has one information set I_2 containing both histories in which player 1 has bet. The game is 2p0s with perfect recall.

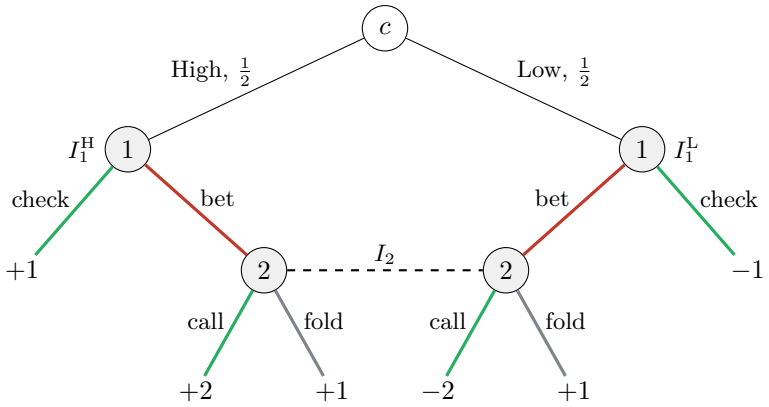


Figure 2.1: The bluffing game of Example 2.2. Chance deals player 1 a high or a low card; player 1 bets or checks; facing a bet, player 2 calls or folds without seeing the card. The dashed line joins the two histories of player 2's single information set I_2. Leaves show player 1's payoff; player 2 receives its negative.

2.1.2 Strategies, best responses and equilibria

A *behavioural strategy* σ_i assigns to each $I \in \mathcal{I}_i$ a distribution $\sigma_i(I) \in \Delta(A(I))$; a *profile* $\sigma = (\sigma_1, \dots, \sigma_n)$ collects one per player, and σ_{-i} denotes all but player i's. A *mixed* strategy is instead a distribution over pure strategies (a choice of action at every information set). For games with perfect recall the two are interchangeable: every mixed strategy has a behavioural strategy that induces the same distribution over terminal histories against every opponent profile, and conversely (Kuhn's theorem [19]). We therefore work with behavioural strategies throughout.

The *reach probability* of a history under σ is the product of the probabilities of the actions along it,

$$\pi^\sigma(h) = \prod_{h' a \sqsubseteq h} \sigma_{P(h')}(h', a),$$

with $\sigma_{P(h')}(h', a)$ meaning $\sigma_{P(h')}(I, a)$ for the information set $I \ni h'$ (and σ_c at chance histories). It factors as $\pi^\sigma(h) = \pi_i^\sigma(h)\,\pi_{-i}^\sigma(h)$, player i's own action probabilities times those of everyone else, chance included. For $h \sqsubseteq z$ we write $\pi^\sigma(h, z)$ for the product over the actions from h to z. Under perfect recall, $\pi_i^\sigma(h)$ is the same for all $h \in I \in \mathcal{I}_i$, and we write it $\pi_i^\sigma(I)$. Expected utility is $u_i(\sigma) = \sum_{z \in Z} \pi^\sigma(z)\, u_i(z)$.

Definition 2.3 (Best response, equilibrium, exploitability)**.** A strategy σ_i^* is a *best response* to σ_{-i} if $u_i(\sigma_i^*, \sigma_{-i}) = \max_{\sigma_i'} u_i(\sigma_i', \sigma_{-i})$. A profile is an *$\epsilon$-equilibrium* if no player can gain more than ϵ by deviating unilaterally, $u_i(\sigma_i', \sigma_{-i}) \le u_i(\sigma) + \epsilon$ for all i and σ_i', and a *Nash equilibrium* if this holds with $\epsilon = 0$. The *exploitability* of σ is the total gain available to unilateral deviations,

$$\varepsilon(\sigma) = \sum_{i \in N} \Big(\max_{\sigma_i'} u_i(\sigma_i', \sigma_{-i}) - u_i(\sigma) \Big) \ge 0.$$

Every finite game has a Nash equilibrium in mixed strategies [25], hence by Kuhn's theorem one in behavioural strategies. A profile with $\varepsilon(\sigma) \le \epsilon$ is an ϵ-equilibrium, and an ϵ-equilibrium has $\varepsilon(\sigma) \le n\epsilon$. In 2p0s games the picture is much stronger.

Theorem 2.4 (Minimax [27])**.** *In a finite 2p0s game,*

$$\max_{\sigma_1} \min_{\sigma_2} u_1(\sigma_1, \sigma_2) = \min_{\sigma_2} \max_{\sigma_1} u_1(\sigma_1, \sigma_2) =: v^*,$$

the value *of the game. A profile is a Nash equilibrium if and only if each strategy guarantees the value to its player, and equilibrium strategies are interchangeable.*

In a 2p0s game $u_1(\sigma) + u_2(\sigma) = 0$, so exploitability no longer involves $u(\sigma)$: $\varepsilon(\sigma) = \max_{\sigma_1'} u_1(\sigma_1', \sigma_2) - \min_{\sigma_2'} u_1(\sigma_1, \sigma_2')$. It is zero exactly at equilibrium, and it bounds how much either strategy can lose against any opponent relative to the value. This is what makes "solving" a 2p0s game well defined, and it is the guarantee that is lost with more players (Section 2.5).

Example 2.5 (Example 2.2 continued)**.** With High, betting is worth at least +1 whatever player 2 does and checking exactly +1, so player 1 bets. Suppose player 1 bets Low with probability p and player 2 calls with probability q. Facing a bet, player 2 holds the posterior $\mathbb{P}(\text{High} \mid \text{bet}) = 1/(1+p)$; calling is worth $(2p-2)/(1+p)$ to player 2 and folding -1, so player 2 is indifferent when $p = \frac{1}{3}$. Player 1 with Low is indifferent between checking (-1) and betting $(1-3q)$ when $q = \frac{2}{3}$. The equilibrium is to bluff a third of the time and call two thirds of the time, and the value to player 1 is $\frac{1}{2}\cdot\frac{5}{3} + \frac{1}{2}\cdot(-1) = \frac{1}{3}$. Two features recur in poker: mixing makes the opponent indifferent, and the bluffing frequency is pinned down by the size of the bet relative to the pot.

2.1.3 Regret and online learning

CFR is built from a simple online learning algorithm. At rounds $t = 1, 2, \ldots$ a learner picks a distribution $x^t \in \Delta(A)$ over a finite set of actions, after which a reward vector $u^t \in \mathbb{R}^A$ with $\max_a u^t(a) - \min_a u^t(a) \le \Delta$ is revealed, possibly chosen by an adversary. Given nonnegative weights w_t, the instantaneous and cumulative *(external) regrets* are

$$r^t(a) = u^t(a) - \langle x^t, u^t \rangle, \qquad R^T(a) = \sum_{t=1}^{T} w_t\, r^t(a) :$$

how much better the learner would have done by always playing a. The plain case is $w_t = 1$. An algorithm is *no-regret* if $\max_a R^T(a)/\sum_t w_t \to 0$ for every reward sequence.

Regret matching plays in proportion to positive cumulative regret,

$$x^{T+1}(a) = \frac{R_+^T(a)}{\sum_{a'} R_+^T(a')} \quad \text{if the denominator is positive, and uniformly otherwise,} \tag{2.1}$$

with $y_+ = \max(y, 0)$. It is Blackwell's approachability strategy [2] for the negative orthant, as observed by Hart and Mas-Colell [15]; see [9] for the general theory. The bound is short to prove.

Theorem 2.6 (Regret matching)**.** *Regret matching guarantees* $\max_a R^T(a) \le \Delta\sqrt{|A|}\,\big(\sum_t w_t^2\big)^{1/2}$. *In particular the average regret is at most* $\Delta\sqrt{|A|/T}$ *when* $w_t = 1$*, and* $O(\Delta\sqrt{|A|/T})$ *when* $w_t = t$.

Proof. First, $\sum_a R_+^T(a)\, r^{T+1}(a) = 0$: this is trivial if $R_+^T = 0$, and otherwise, writing $\Sigma = \sum_a R_+^T(a)$ so that $x^{T+1} = R_+^T/\Sigma$, it equals $\sum_a R_+^T(a) u^{T+1}(a) - \Sigma\,\langle x^{T+1}, u^{T+1}\rangle = 0$. Second, $\big((y+z)_+\big)^2 \le (y_+ + z)^2$ for real y, z (if $y + z > 0$ then $0 < y + z \le y_+ + z$). Applying it componentwise,

$$\begin{aligned} \|R_+^{T+1}\|^2 \le \|R_+^T + w_{T+1} r^{T+1}\|^2 &= \|R_+^T\|^2 + 2w_{T+1}\langle R_+^T, r^{T+1}\rangle + w_{T+1}^2 \|r^{T+1}\|^2 \\ &\le \|R_+^T\|^2 + w_{T+1}^2 |A| \Delta^2, \end{aligned}$$

since $|r^t(a)| \le \Delta$. By induction $\|R_+^T\|^2 \le |A|\Delta^2 \sum_t w_t^2$, and $\max_a R^T(a) \le \|R_+^T\|$. For $w_t = t$, $(\sum_t t^2)^{1/2}/\sum_t t \sim (2/\sqrt{3})\, T^{-1/2}$. □

No-regret learning yields equilibria in 2p0s games by self-play. For extensive-form games one must be careful about what "average strategy" means: averaging behavioural strategies information set by information set does not average the induced play. The right average weights each iterate by the player's own probability of reaching the information set,

$$\bar\sigma_i(I, a) = \frac{\sum_t w_t\, \pi_i^{\sigma^t}(I)\, \sigma_i^t(I, a)}{\sum_t w_t\, \pi_i^{\sigma^t}(I)}. \tag{2.2}$$

By induction along h, $\pi_i^{\bar\sigma}(h)$ is then the w-weighted average of the $\pi_i^{\sigma^t}(h)$, and since utility is linear in a player's reach probabilities, $u_j(\bar\sigma_i, \sigma_{-i}) = \sum_t w_t u_j(\sigma_i^t, \sigma_{-i}) / \sum_t w_t$ for every σ_{-i} and every j.

Proposition 2.7 (Folk theorem for 2p0s)**.** *In a 2p0s game, let both players choose σ_i^t so that their average regret against the sequence of opponent strategies is at most ϵ, $\max_{\sigma_i'} \sum_t w_t\big(u_i(\sigma_i', \sigma_{-i}^t) - u_i(\sigma^t)\big) \le \epsilon \sum_t w_t$. Then $\varepsilon(\bar\sigma) \le 2\epsilon$ for the average profile (2.2); in particular $\bar\sigma$ is a 2ϵ-equilibrium.*

Proof. Write $\mathbb{E}_t$ for the w-weighted average over t. By linearity and the regret bound for player 1, for any σ_1', $u_1(\sigma_1', \bar\sigma_2) = \mathbb{E}_t\, u_1(\sigma_1', \sigma_2^t) \le \mathbb{E}_t\, u_1(\sigma^t) + \epsilon$. Likewise for player 2, using $u_2 = -u_1$: $u_1(\bar\sigma_1, \sigma_2') = \mathbb{E}_t\, u_1(\sigma_1^t, \sigma_2') \ge \mathbb{E}_t\, u_1(\sigma^t) - \epsilon$. Subtracting, $\max_{\sigma_1'} u_1(\sigma_1', \bar\sigma_2) - \min_{\sigma_2'} u_1(\bar\sigma_1, \sigma_2') \le 2\epsilon$, which is $\varepsilon(\bar\sigma)$. Since $u_1(\bar\sigma)$ lies between the two terms, neither player gains more than 2ϵ by deviating. □

2.2 Counterfactual regret minimisation

Running regret matching on the pure strategies of an extensive-form game is hopeless: there are exponentially many. CFR [35] instead runs one regret minimiser at every information set, on a local notion of value. For $I \in \mathcal{I}_i$, the *counterfactual value*

$$v_i^\sigma(I) = \sum_{h \in I} \pi_{-i}^\sigma(h) \sum_{z \sqsupseteq h} \pi^\sigma(h, z)\, u_i(z) \tag{2.3}$$

is player i's expected payoff from I onwards, weighted by the probability that chance and the other players bring play to each $h \in I$, as if i had played to reach I. We write $v_i^\sigma(I, a)$ for the same quantity with action a forced at I, so that $v_i^\sigma(I) = \sum_a \sigma_i(I, a)\, v_i^\sigma(I, a)$. The cumulative *counterfactual regret* after T iterations is

$$R^T(I, a) = \sum_{t=1}^{T} w_t\big(v_i^{\sigma^t}(I, a) - v_i^{\sigma^t}(I)\big), \tag{2.4}$$

and CFR sets $\sigma^{T+1}(I)$ by regret matching (2.1) on $R^T(I, \cdot)$. Each information set is thus an instance of Section 2.1.3 with reward vector $v_i^{\sigma^t}(I, \cdot)$, whose range is at most $\Delta_i = \max_z u_i(z) - \min_z u_i(z)$ because the weights π_{-i} sum to at most one.

Theorem 2.8 (Zinkevich et al. [35])**.** *Player i's overall regret is bounded by the sum of the positive counterfactual regrets, $\max_{\sigma_i'} \sum_t w_t\big(u_i(\sigma_i', \sigma_{-i}^t) - u_i(\sigma^t)\big) \le \sum_{I \in \mathcal{I}_i} \max_a R_+^T(I, a)$. With regret matching at every information set and $w_t = 1$, the average overall regret is therefore at most $\Delta_i\, |\mathcal{I}_i| \sqrt{|A_i|} / \sqrt{T}$, where $|A_i|$ is the largest number of actions at any of i's information sets.*

Proof sketch. Fix a deviation σ_i'. The gain from it decomposes over i's information sets in a top-down order: switching from σ_i^t to σ_i' at I while keeping σ_i' below changes the counterfactual value at I by a term bounded by the local regret there, plus the gains at the information sets that follow, each weighted by $\sigma_i'(I, a) \le 1$. Perfect recall makes the counterfactual values of successive information sets compose, because π_{-i} does not depend on i's own choices. Unrolling the recursion gives the first bound, and Theorem 2.6 at each information set gives the second. □

Combining Theorem 2.8 with Proposition 2.7, in a 2p0s game the average profile of CFR, weighted by each player's own reach as in (2.2), is an ϵ-equilibrium with $\epsilon = O(1/\sqrt{T})$. The argument goes through unchanged for nonnegative weights w_t, with the factor of Theorem 2.6 in place of $\sqrt{T}$; this is how the linear weights of Section 2.4 keep the guarantee.

2.3 Monte Carlo CFR

Each CFR iteration evaluates (2.3) exactly, which means a pass over the whole tree including every deal of the cards. With six players there are $52!/(2^6\,40!)$ deals, so this is out of the question. Monte Carlo CFR (MCCFR) [20] replaces the counterfactual values by unbiased estimates computed on a sampled part of the tree. If terminal z is sampled with probability $q(z) > 0$, the sampled counterfactual value

$$\tilde{v}_i^{\sigma}(I) = \sum_{h \in I} \sum_{\substack{z \sqsupseteq h \\ z \text{ sampled}}} \frac{\pi_{-i}^{\sigma}(h)\,\pi^{\sigma}(h,z)}{q(z)}\,u_i(z)$$

is unbiased for $v_i^{\sigma}(I)$, and regret matching on sampled regrets keeps a regret bound of the same order with high probability, with constants that grow with the variance of the estimates [20]. Two sampling schemes are standard. *Outcome sampling* follows a single path from the root to one terminal. *External sampling*, which we use, singles out a *traverser* i per iteration: at chance and opponent histories it samples one action from the actual distribution σ^t, and at the traverser's histories it enumerates every action. The importance weight then disappears.

Proposition 2.9 (External sampling has importance weight one)**.** *In one external-sampling pass for traverser i, at most one history of each $I \in \mathcal{I}_i$ is visited. Let $\tilde{v}(I,a)$ be the value the recursion of Algorithm 1 returns from the child ha of the visited $h \in I$ (zero if no $h \in I$ is visited), and $\tilde{v}(I) = \sum_a \sigma^t(I,a)\,\tilde{v}(I,a)$. Then $\mathbb{E}\,\tilde{v}(I,a) = v_i^{\sigma^t}(I,a)$, so the sampled regret increment $\tilde{v}(I,a) - \tilde{v}(I)$ is unbiased for $v_i^{\sigma^t}(I,a) - v_i^{\sigma^t}(I)$.*

Proof. Two histories of I agree on i's own actions (perfect recall), so they first differ at a chance or opponent history, where only one action is sampled; hence at most one is visited. A history h is visited exactly when every chance and opponent action on the way to it is sampled, which has probability $\pi_{-i}^{\sigma^t}(h)$ since the traverser's actions are all enumerated. Given that, the recursion returns

$$\tilde{v}(I,a) = \sum_{z \sqsupseteq ha} \pi_i^{\sigma^t}(ha,z)\,u_i(z)\,\mathbb{1}[\text{every non-}i\text{ action from } ha \text{ to } z \text{ is sampled}]\,:$$

the traverser's actions are averaged with their own probabilities, and the others' are realised by sampling. The indicator has conditional expectation $\pi_{-i}^{\sigma^t}(ha,z)$. Summing over the disjoint events $\{h \text{ visited}\}$, $h \in I$,

$$\mathbb{E}\,\tilde{v}(I,a) = \sum_{h \in I} \pi_{-i}^{\sigma^t}(h) \sum_{z \sqsupseteq ha} \pi_i^{\sigma^t}(ha,z)\,\pi_{-i}^{\sigma^t}(ha,z)\,u_i(z) = v_i^{\sigma^t}(I,a).$$

In the notation above $q(z) = \pi_{-i}^{\sigma^t}(z)$, so each sampled term carries the weight $\pi_{-i}(h)\pi(h,z)/q(z) = \pi_i(h,z)$, a probability of the traverser's own play: there is no importance weight. Linearity gives the statement for the regret increment. □

The proposition assumes exact terminal utilities. Ours are Monte Carlo estimates, unbiased at all-ins but slightly biased at flop terminals (Proposition 2.12), so there the unbiasedness is approximate.

Averaging at opponent histories. External sampling never computes $\pi_p^{\sigma^t}(I)$, which (2.2) needs. A common remedy, which we follow, adds $w_t\sigma^t(I)$ to a strategy sum $S(I)$ whenever $I \in \mathcal{I}_p$ is visited during another player's traversal, and outputs $\bar{\sigma}(I) = S(I)/\|S(I)\|_1$. In a pass with traverser $i \neq p$, a history $h \in I$ is visited with probability $\pi_{-i}^{\sigma^t}(h) = \pi_p^{\sigma^t}(I)\,\pi_{-\{i,p\}}^{\sigma^t}(h)$, so the expected increment is

$$w_t\,\pi_p^{\sigma^t}(I)\,c_t(I)\,\sigma^t(I), \qquad c_t(I) = \sum_{h \in I} \pi_{-\{i,p\}}^{\sigma^t}(h),$$

Algorithm 1 External-sampling MCCFR with block-linear weights. R and S are the regret and strategy-sum tables, initialised to zero, and N is the block length.

function TRAVERSE(h, i, w)
  if $h \in Z$ **then**
    return $\tilde{u}_i(h)$ ▷ sampled payoff, Section 2.6
  $p \leftarrow P(h)$; $I \leftarrow$ the information set of p containing h
  $\sigma(I, \cdot) \leftarrow$ regret matching (2.1) on $R(I, \cdot)$
  if $p = i$ **then**
    for $a \in A(I)$ **do**
      $\tilde{v}(a) \leftarrow$ TRAVERSE(ha, i, w)
    $\tilde{v} \leftarrow \sum_a \sigma(I, a)\, \tilde{v}(a)$
    for $a \in A(I)$ **do**
      $R(I, a) \leftarrow R(I, a) + w\,(\tilde{v}(a) - \tilde{v})$
    return $\tilde{v}$
  else
    $S(I, \cdot) \leftarrow S(I, \cdot) + w\,\sigma(I, \cdot)$ ▷ averaging at opponent histories
    sample $a \sim \sigma(I, \cdot)$
    return TRAVERSE(ha, i, w)

for $t = 1, \ldots, T$ **do**
  sample a deal ω of six hands ▷ chance, from its own distribution
  $i \leftarrow t \bmod n$; $w \leftarrow \lfloor (t-1)/N \rfloor + 1$ ▷ the block number
  TRAVERSE(ω, i, w)
return $\bar{\sigma}(I) = S(I)/\|S(I)\|_1$ for every I (regret matching where $S(I) = 0$)

averaged over the traverser. This is the textbook weight times $c_t(I)$, the probability that chance and the players other than p and the traverser lead to I. If $c_t(I)$ were constant in t it would cancel on normalisation. It changes only through the other players' strategies, so it drifts more and more slowly as they settle. An information set whose strategy sum stays zero has never been visited during another player's traversal, and we output its regret-matching strategy instead.

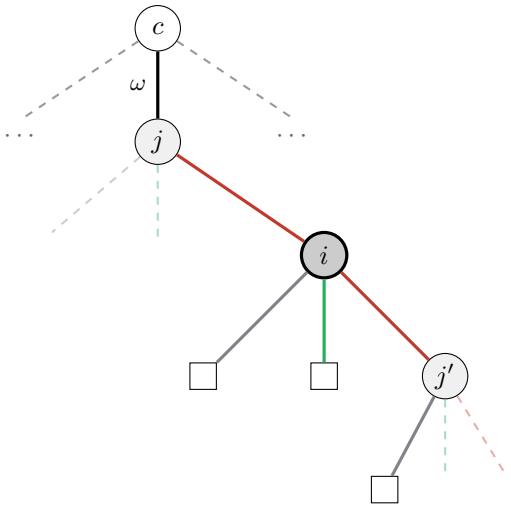
Algorithm 1 is our traversal with the linear block weights of Section 2.4, and Figure 2.2 shows one pass. The root is the chance node of the deal: six hands are drawn once per iteration, and the remaining chance (the boards) is handled inside the payoff estimate (Section 2.6). The traverser cycles through the seats.

2.4 Weighted averaging and other variants

Vanilla CFR weights every iteration equally, so the early iterations, played against nearly uniform opponents, weigh on the average for a long time. Several variants change this. CFR+ [33] floors the stored regrets at zero after every update (*regret matching*$^+$), alternates the players' updates and weights the average linearly; it was the algorithm behind the essential solution of heads-up limit hold'em [3]. Brown and Sandholm's Discounted CFR [6] multiplies the accumulated positive and negative regrets by $t^\alpha/(t^\alpha + 1)$ and $t^\beta/(t^\beta + 1)$, and the strategy sum by $(t/(t+1))^\gamma$, after iteration t. Its simplest case, *Linear CFR*, weights the contributions of iteration t to both tables by t. We apply Linear CFR a block at a time.

Proposition 2.10 (Block discounting)**.** *Split the iterations into blocks of length N, block j being $(j-1)N < t \le jN$.*

(a) If at the end of each block k both tables are multiplied by $k/(k+1)$, then after K blocks the



Chance: sample one deal ω.

Player $j \neq i$: sample $a \sim \sigma^t(I)$ and add $w_t \sigma^t(I)$ to $S(I)$.

Traverser i: explore every $a \in A(I)$ and add $w_t\big(\tilde{v}(I,a) - \tilde{v}(I)\big)$ to $R(I,a)$.

Terminal: return the sampled payoff $\tilde{u}_i$ (Section 2.6).

Figure 2.2: One external-sampling pass (Algorithm 1). Solid edges are explored; faded dashed ones are not. At the traverser's node every action (fold, call, raise) is explored; at chance and at the other players' nodes one action is drawn from its actual probability, which is why the importance weight is one (Proposition 2.9).

increments of block j carry the weight $\prod_{k=j}^{K} \frac{k}{k+1} = \frac{j}{K+1}$, proportional to j.

(b) If instead each increment made during block j is multiplied by j and the tables are never rescaled, then at every iteration of block k the tables equal k times those of (a), and after the last block $K+1$ times. Both schemes produce the same strategies σ^t and the same average $\bar{\sigma}$.

Proof. (a) The product telescopes. (b) Let Δ_j be the sum of the raw increments of block j and Δ'_k those made so far in block k. By (a), during block k the rescaled table holds $D = \sum_{j<k} \frac{j}{k}\Delta_j + \Delta'_k$, and the lazy table holds $\sum_{j<k} j\,\Delta_j + k\,\Delta'_k = kD$. Regret matching (2.1) and the normalisation $S(I)/\|S(I)\|_1$ are invariant under multiplying their input by a positive constant, so the two schemes play identical strategies at every iteration (given the same random numbers) and output the same average. By Theorem 2.6 with $w_t = \lceil t/N \rceil$, the regret bound keeps the order $O(1/\sqrt{T})$. □

The lazy form (b) is what Algorithm 1 does through w. It needs no pass over the tables and no synchronisation between threads (Section 3.4).

Why we do not floor regrets. Flooring interacts badly with sampled regrets. Given the current table entry Q and an unbiased but noisy increment $\tilde{r}$ with mean r, Jensen's inequality for the convex map $y \mapsto y_+$ gives $\mathbb{E}\,(Q+\tilde{r})_+ \geq (Q+r)_+$: the floored update is biased upwards. The bias is largest for actions whose true regret hovers near zero, which are exactly the ones an equilibrium mixes or barely rejects. Without the floor, negative regret acts as a memory: an action that was bad for a long time must earn its way back before it is played again, and noise in either direction averages out. Linear and Discounted CFR keep negative regrets (DCFR merely discounts them), and that is the family we stay in.

2.5 More than two players

Theorems 2.6 and 2.8 are statements about one learner against an arbitrary sequence of rewards, so they hold for each player in any game: CFR still drives every player's average regret to zero. What fails beyond 2p0s is Proposition 2.7, whose proof used $u_2 = -u_1$ to tie the two players'

guarantees together. Nothing replaces it. What no-regret self-play does deliver in general is a weaker, correlated notion.

Proposition 2.11. *If every player's average regret over T iterations is at most ϵ, the joint distribution μ that draws t with probability $w_t / \sum_s w_s$ and plays σ^t is an ϵ-coarse correlated equilibrium: for every i and σ_i', $\mathbb{E}_\mu\, u_i(\sigma_i', \sigma_{-i}^t) \le \mathbb{E}_\mu\, u_i(\sigma^t) + \epsilon$.*

Proof. Multiplied by $\sum_t w_t$, the displayed inequality is the regret bound. □

A coarse correlated equilibrium is a mixture of profiles, not a profile; internal-regret dynamics give the stronger correlated equilibria [15], and Farina et al. [11] study coarse correlation in extensive-form games. The profile $\bar{\sigma}$ of per-player averages that CFR outputs is in general no equilibrium of any kind, and even Nash equilibria need not be unique or interchangeable, so "the" equilibrium strategy of a seat is not well defined. The practical evidence is nonetheless strong. Pluribus [7] computed the blueprint strategy for six-player no-limit hold'em with Monte Carlo CFR and beat professional players, and Gibson [13] documents CFR agents performing well in three-player poker.

Rake makes the game negative-sum, $\sum_i u_i(z) = -\operatorname{rake}(z)$ (Section 2.6.2). Adding the house as a passive player with utility $\operatorname{rake}(z)$ restores zero-sum, but with seven players, so it buys no guarantee; even heads-up with rake is outside Proposition 2.7, which is why our exact checks run with rake off. With no theorem to lean on, we check convergence empirically (Section 4.2).

2.6 Estimators used in this work

2.6.1 Equity

At a terminal reached before the flop, let L be the set of players still in the hand. Given the hole cards, a board B of five cards is drawn uniformly from the remaining deck (Section 3.2 says exactly which cards are removed), and $W(B) \subseteq L$ is the set of players with the best seven-card hand on B. The *equity* of $i \in L$ is the expected share of the pot i would win at showdown with no further betting,

$$\mathrm{eq}_i = \mathbb{E}_B\left[\frac{\mathbb{1}[i \in W(B)]}{|W(B)|}\right], \qquad \sum_{i \in L} \mathrm{eq}_i = 1.$$

Heads-up, eq_i is an exact sum over the $\binom{48}{5}$ boards, which we precompute for every pair of hands. With three or more players we use the Monte Carlo estimator over K independent boards,

$$\widehat{\mathrm{eq}}_i = \frac{1}{K}\sum_{k=1}^{K} \frac{\mathbb{1}[i \in W(B_k)]}{|W(B_k)|},$$

which is unbiased with variance at most $1/(4K)$, since each term lies in $[0, 1]$. At an all-in we use $K = 1$ and pay the actual pot share on that board. The payoff estimate is then unbiased, and the noise averages out over iterations.

2.6.2 Realization-adjusted pot shares

Without an all-in, the hand would continue to the flop, and raw equity is a poor proxy for what a hand wins there: position, initiative and playability let some hands win more than their share and others less. We model this with a *realization factor* $\rho_i > 0$ per live player: an abstract positive function of the player's hand class, role (position, caller or raiser) and the depth of the pot, with parameters θ. Its concrete form is the subject of Section 3.3. Pot shares are realization-adjusted equities renormalised over L,

$$s_i = \frac{\rho_i\, \mathrm{eq}_i}{\sum_{j \in L} \rho_j\, \mathrm{eq}_j}, \qquad u_i(z) = s_i\,(\operatorname{pot}(z) - \operatorname{rake}(z)) - C_i(z), \tag{2.5}$$

with $s_i = 0$ for $i \notin L$, where $C_i(z)$ is player i's contribution to the pot and $\mathrm{rake}(z)$ the rake. Since $\sum_i C_i = \mathrm{pot}$, the utilities sum to $-\mathrm{rake}(z)$ at every terminal and for every board. The same payoff form covers the other terminals, with $s_i = 1$ for the last player when everyone else folds and $s_i = \mathrm{eq}_i$ at an all-in.

The shares have three properties we use.

(i) *Scale invariance.* Multiplying every ρ_j by the same $c > 0$ leaves every s_i unchanged. Only ratios of realization factors matter, so one parameter can be fixed by convention.

(ii) *Consistency.* If all ρ_j are equal, $s_i = \mathrm{eq}_i$.

(iii) *One board is not enough.* With $K = 1$ and no tie, $\widehat{\mathrm{eq}}_j = \mathbb{1}[j \text{ wins}]$, and (2.5) collapses to $s_i = \mathbb{1}[i \text{ wins}]$ whatever the ρ_j: the realization model has no effect. A flop terminal therefore needs a genuine equity estimate, exact heads-up and with K boards multiway.

Proposition 2.12 (Bias of the multiway share)**.** *Let $f(e) = \rho_i e_i / \sum_j \rho_j e_j$, and let $\widehat{\mathrm{eq}}$ average K independent single-board share vectors with covariance Σ_1. Then $\mathbb{E} f(\widehat{\mathrm{eq}}) = f(\mathrm{eq}) + \frac{1}{2K}\operatorname{tr}(\nabla^2 f(\mathrm{eq})\,\Sigma_1) + o(1/K)$.*

Proof sketch. f takes values in $[0,1]$ and is smooth wherever $\sum_j \rho_j e_j > 0$, in particular near eq. A second-order Taylor expansion about eq, with $\mathbb{E}(\widehat{\mathrm{eq}} - \mathrm{eq}) = 0$ and $\mathrm{Cov}(\widehat{\mathrm{eq}}) = \Sigma_1/K$, gives the leading terms. The remainder is $o(1/K)$ because $\widehat{\mathrm{eq}}$ concentrates around eq exponentially fast (Hoeffding's inequality) and f is bounded; this is the delta method [28]. □

The share is a ratio of noisy estimates, so it is biased, but only at order $1/K$. Heads-up the exact table removes the bias altogether. We choose K by checking that the solution stops moving as K grows.

2.6.3 Self-normalised importance sampling

The trainer (Section 3.9) and the fixed-strategy fit (Section 3.8) need the expected value of each action for a given hand of one player, the *hero*, at a given spot, assuming everyone plays a fixed strategy from there on. The spot is reached through a history h, which makes some deals of the opponents' cards more likely than others. Deals are easy to draw uniformly from the cards the hero does not hold (the proposal q), but the target is the conditional distribution given h,

$$p(\omega) = \frac{q(\omega)\,\ell(\omega)}{Z}, \qquad \ell(\omega) = \prod_{h'a' \sqsubseteq h,\ P(h') \neq \mathrm{hero}} \sigma\big(h', b(\omega_{P(h')})\big)(a'), \qquad Z = \mathbb{E}_q\,\ell,$$

where ℓ is the probability that the other players' strategies produce the observed actions with the cards ω, and $b(\omega_p)$ is the class of player p's hand. Z is unknown, which is exactly the setting of *self-normalised importance sampling* [28]. With deals $\omega_1, \dots, \omega_N \sim q$, weights $\ell_n = \ell(\omega_n)$, and a playout payoff $u_n(a)$ for action a on deal n, the estimator and its standard error are

$$\widehat{\mathrm{EV}}(a) = \frac{\sum_n \ell_n\, u_n(a)}{\sum_n \ell_n}, \qquad \widehat{\mathrm{SE}}(a) = \frac{\big(\sum_n \ell_n^2 \big(u_n(a) - \widehat{\mathrm{EV}}(a)\big)^2\big)^{1/2}}{\sum_n \ell_n}. \tag{2.6}$$

By the strong law of large numbers, numerator and denominator divided by N converge to $\mathbb{E}_q[\ell u] = Z\,\mathbb{E}_p[u]$ and $\mathbb{E}_q[\ell] = Z$, so the estimator is consistent. Being a ratio, it is biased at order $1/N$. The delta method gives $\sqrt{N}(\widehat{\mathrm{EV}} - \mathbb{E}_p u) \Rightarrow \mathcal{N}\big(0, \mathbb{E}_q[\ell^2(u - \mathbb{E}_p u)^2]/Z^2\big)$, and $\widehat{\mathrm{SE}}^2$ is the plug-in estimate of that variance over N.

Precision is governed by the weights, not by N. Kish's *effective sample size* [18]

$$N_{\mathrm{eff}} = \frac{\big(\sum_n \ell_n\big)^2}{\sum_n \ell_n^2} \in [1, N]$$

is the number of equally weighted draws that would give about the same variance, $\operatorname{Var}\widehat{\mathrm{EV}} \approx \operatorname{Var}_p(u)/N_{\mathrm{eff}}$, when the weights are not strongly correlated with the payoffs. A rare history leaves most deals with tiny weight and a small N_{eff}, so we deal until N_{eff} reaches a target, or a cap on N.

Every action is played out on the same deals (*common random numbers*). Then $\operatorname{Var}\big(\widehat{\mathrm{EV}}(a) - \widehat{\mathrm{EV}}(a')\big) = \operatorname{Var}\widehat{\mathrm{EV}}(a) + \operatorname{Var}\widehat{\mathrm{EV}}(a') - 2\operatorname{Cov}$, and the covariance is large and positive because the deal drives most of the payoff. Differences between actions, which are what decisions depend on, are therefore much more precise than the levels. Their standard error is (2.6) applied to $u_n(a) - u_n(a')$.

2.7 Fitting a payoff model to observed play

The realization parameters θ are not observable, so we choose them so that the solved strategies resemble a set of reference charts (Section 3.6). This section sets out the two objectives we use and how they relate to models of boundedly rational play.

Definition 2.13 (Weighted total-variation distance)**.** At the node h_s of a reference spot s, let $p = P(h_s)$, and give each hand class k the weight $w_k = n_k\,\pi_p^{\bar\sigma}(h_s, k)$: its number of combinations n_k times the probability that p's own earlier actions under $\bar\sigma$ bring class k to h_s. The distance between the strategy and the chart σ_s^{ref} at s is

$$D_s(\bar\sigma) = \frac{\sum_k w_k\ \mathrm{TV}\left(\bar\sigma(h_s,k),\,\sigma_s^{\mathrm{ref}}(k)\right)}{\sum_k w_k} \in [0,1], \qquad \mathrm{TV}(x,y) = \tfrac{1}{2}\|x-y\|_1.$$

Since $\mathrm{TV}(x,y)$ is the smallest probability, over couplings, that draws from x and y differ, D_s is the probability that the strategy and the chart choose different actions for a class drawn in proportion to w_k, under the best coupling for each class. Classes the strategy never brings to the spot do not count, and frequent classes count most. The reference spots are grouped into categories c (opens, facing an open, facing a 3-bet, and so on). The objective averages the spots within each category and then the categories, so that a category with many spots does not dominate:

$$J(\theta) = \frac{1}{|\mathcal{C}|}\sum_{c\in\mathcal{C}}\frac{1}{|\mathcal{S}_c|}\sum_{s\in\mathcal{S}_c} D_s(\bar\sigma_\theta) + \mu\sum_p \operatorname{dist}\left(o_p(\bar\sigma_\theta), [l_p, u_p]\right),$$

where $\bar\sigma_\theta$ is the solved strategy under θ, and the second term penalises open frequencies o_p that leave sanity bands $[l_p, u_p]$ (Section 3.7).

Each evaluation of J is a full solve, so J is expensive, noisy and offers no gradient. Pattern search [16] needs only comparisons: it tries a step up and down each coordinate, moves when a step improves by more than a few standard errors of the difference, and halves the steps when none does (Section 3.7). Every trial costs a solve, which caps the number of parameters that can be tuned.

Quantal response and the likelihood view. A cheaper objective treats the charts as noisy best responses. Fix every player's strategy, so that each action a of class k at spot s has an expected value $\mathrm{EV}_\theta(s,k,a)$, and model play by the *logit response*

$$q_\theta(a \mid s,k) = \operatorname{softmax}\left(\mathrm{EV}_\theta(s,k,\cdot)/\tau\right)_a = \frac{\exp(\mathrm{EV}_\theta(s,k,a)/\tau)}{\sum_{a'}\exp(\mathrm{EV}_\theta(s,k,a')/\tau)},$$

which tends to a best response as $\tau \to 0$ and to uniform play as $\tau \to \infty$. McKelvey and Palfrey [23] define the quantal response equilibrium as a profile in which every player plays the logit response to the others; holding the others fixed, as we do, is a one-sided version with no fixed point to compute. If the chart mix $p_{s,k}$ is read as observed action frequencies, the weighted

log-likelihood of the charts is $\sum_{s,k} \omega_{s,k} \sum_a p_{s,k}(a) \log q_\theta(a \mid s,k)$. Maximising it is the same as minimising

$$\mathcal{L}(\theta) = \sum_{s,k} \omega_{s,k} \; \mathrm{KL}\left(p_{s,k} \,\|\, q_\theta(\cdot \mid s,k)\right), \qquad \frac{\partial \; \mathrm{KL}(p_{s,k} \,\|\, q_\theta)}{\partial \, \mathrm{EV}_\theta(s,k,a)} = \frac{q_\theta(a \mid s,k) - p_{s,k}(a)}{\tau},$$

since the two differ by the charts' entropy, which does not depend on θ. The weights $\omega_{s,k}$ are those of Definition 2.13 with the category balancing of J. An L^2 penalty pulling θ towards a start point is a Gaussian prior, which makes the minimiser a maximum a posteriori estimate. A complementary diagnostic is the chart's *regret* $\max_a \mathrm{EV}_\theta(s,k,a) - \sum_a p_{s,k}(a) \, \mathrm{EV}_\theta(s,k,a) \geq 0$: how much the chart gives up against the best action, in big blinds.

Choosing the payoffs of a game so that observed play is (quantally) rational is *inverse game theory*. Ling et al. [22] learn game parameters end to end by differentiating through the quantal response equilibrium of normal- and extensive-form games. Our fixed-strategy fit (Section 3.8) is the simpler variant in which the other players' strategies stay fixed, so the loss and its gradient need no equilibrium computation. The price is that the fit cannot see how the other players would respond to a change in θ (Section 4.5).

Chapter 3

Methodology

This chapter turns the mathematics of Chapter 2 into a working system. Figure 3.1 shows the pipeline: an exact heads-up equity table is computed once; a solve from scratch produces the *Scratch* strategy; its realization parameters are fitted to reference charts; a second solve with the charts locked produces the *Anchored* strategy; and the trainer plays both.

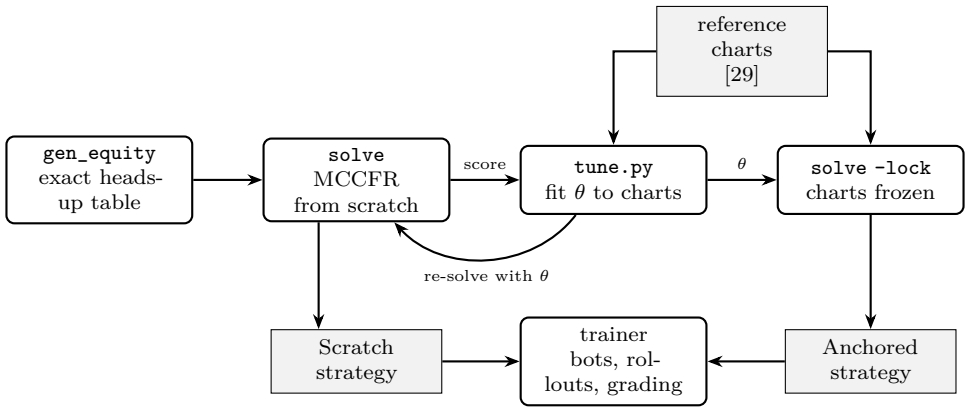


Figure 3.1: The pipeline that produces the two shipped strategies. The tuning loop re-solves from scratch for every candidate θ.

3.1 Encoding the game

Rules and tree. All rules of Section 1.3 live in one configuration structure (stack, blinds, raise multipliers, tick, rake, and whether a cold caller may call a squeeze), and a rules engine applies them to a hand state. The betting tree is built once by walking the engine from the empty history: each decision node stores the actor, a mask of legal actions and a child index per action, with -1 marking a terminal. Because all stacks are equal, an all-in is always a plain showdown with no side pots.

Terminals. Everything a terminal's payoff depends on, apart from the cards, is fixed by its history: each player's contribution C_i, the pot after rake, the stack-to-pot ratio SPR at the flop, which players are live, who acts last postflop, who made the last raise, whether the pot was 3-bet or more, and whether an all-in was called. These are precomputed once per terminal edge, so the solver traverses the tree by node index and never touches the rules engine in its inner loop.

Information sets and deals. An information set is a pair (decision node, hand class), and strategies and regrets are stored per pair (Section 1.4). Each iteration deals six real combos from one shuffled deck, so blockers shape the opponents' hands exactly, while the strategy lookup uses

only the class. Smaller games are solved on the same tree by forcing the early seats to fold at the root: `-seats 2` gives SB versus BB, which we use as a correctness check in Section 4.1.

3.2 Equity computation

Heads-up: an exact table. For two disjoint combos the equity is a sum over all $\binom{48}{5} = 1{,}712{,}304$ boards (Section 2.6.1). We precompute it for every ordered pair of disjoint combos. Matchups that differ only by a relabelling of suits, or by swapping the two sides, have equal (or complementary) equity, so each canonical matchup is evaluated once and the result is expanded into a 1326×1326 table of 32-bit floats (7.0 MB). Lookups are then exact and free. One simplification is that the table ignores the cards of folded players; we return to this in Section 4.6.

Multiway: sampled boards. With three or more players we draw $K = 100$ boards from the deck minus all twelve hole cards and average each player's share of the pot (Section 2.6.1). This is the solver's hot loop, so the seven-card evaluator is specialised. Each card is given a 64-bit key that packs a base-5 digit for its rank and a 4-bit count for its suit. A seven-card hand's key is then the sum of its cards' keys, and a board's key is computed once and shared by every player. Hands without a flush are ranked by a table lookup on the two rank fields, built at start-up from PokerHandEvaluator [21]; the rare flushes fall back to the library itself. The results are bit-identical to calling the library directly.

All-ins. A called all-in uses a single sampled board and pays the actual pot share on it. This estimator is unbiased for the expected payoff, and the noise averages out over iterations.

3.3 The realization model

At a terminal where the flop would be dealt, player i receives the share $s_i = \rho_i \mathrm{eq}_i / \sum_j \rho_j \mathrm{eq}_j$ of the pot after rake (Equation (2.5)). This section defines the realization factor $\rho_i > 0$. It is built from features of the hand, the player's *role* and the depth of the pot. Fold equity and blockers are *not* modelled here: the solver deals real cards and plays the preflop betting out exactly, so both are already accounted for, and a blocker term would count them twice.

Depth. With SPR the stack behind divided by the pot at the flop, and a parameter $\kappa \geq 0$, let

$$\gamma = \frac{\mathrm{SPR}}{\mathrm{SPR} + \kappa} \in (0, 1]. \tag{3.1}$$

γ is close to 1 in deep single-raised pots and falls in 3-bet and 4-bet pots; at $\mathrm{SPR} = \kappa$ it equals $\frac{1}{2}$, so κ reads as the 'half-life' SPR (Figure 3.2). Terms whose value comes from playing future streets are scaled by γ, so they fade when little money is left behind.

Roles. Call player i an *out-of-position caller* if i neither acts last postflop among the live players nor made the last raise, so has neither position nor initiative, and an *in-position caller* if i acts last but did not make the last raise. Call the pot *re-raised* ($\mathbb{1}_{\mathrm{rr}} = 1$) if it was 3-bet or more; this is read from the betting history, not from SPR. The position factor is

$$\mathrm{pos}_i = \begin{cases} \alpha_{\mathrm{ip}}\, \chi_i & i \text{ acts last postflop,} \\ 1 - (1 - \alpha_{\mathrm{oop}})\, \gamma_i' & \text{otherwise,} \end{cases} \tag{3.2}$$

where $\chi_i = \chi$ for an in-position caller in a re-raised pot and 1 otherwise, and $\gamma_i' = \xi\gamma + (1 - \xi)$ for an out-of-position caller and γ otherwise. The out-of-position penalty $1 - \alpha_{\mathrm{oop}}$ shrinks with

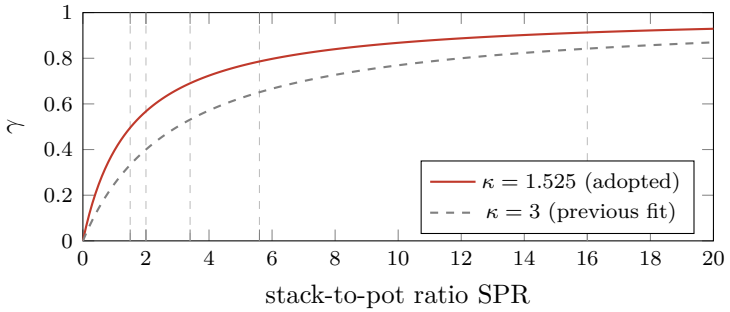


Figure 3.2: The depth fade $\gamma(\mathrm{SPR})$. Dashed grid lines mark typical heads-up flop SPRs: about 16 in a single-raised pot, 3.4 to 5.6 in a 3-bet pot and 1.5 to 2 in a 4-bet pot.

depth, since with little behind there is little to be outplayed on; $\xi \in [0,1]$ controls how much of that shrink an out-of-position caller gets. χ captures that calling a 3-bet in position, with a capped range into a small SPR against the aggressor's initiative, is worth less than position in a single-raised pot.

Hand features. For a hand with ranks $r_1 \geq r_2$ (2 is 0, A is 12):

- $\mathrm{st}(r_1, r_2)$, the *connectivity*: the number of five-card straights, from the wheel A-2-3-4-5 (the ace playing low) up to T-J-Q-K-A, that contain both ranks; 0 for pairs. For example st = 4 for 98, 2 for KQ, 1 for AK and A5, and 0 for K4.
- $\mathrm{dom}(r_1, r_2)$, the *domination*: for offsuit non-pairs with $r_1 \geq$ T, $\min(r_1 - 1 - r_2,\, 4)$, else 0. It measures how often the hand makes top pair with a weak kicker against better versions of itself: KQo has 0, KJo 1, AJo 2, and K6o is capped at 4.
- Indicators $\mathbb{1}_{\mathrm{suited}}$, $\mathbb{1}_{\mathrm{pair}}$, $\mathbb{1}_{\mathrm{off}}$ (offsuit non-pair) and $\mathbb{1}_{\mathrm{raiser}}$ (i made the last raise).

The hand factor is

$$H_i = 1 + \left(\phi\,\mathbb{1}_{\mathrm{suited}} + \eta\,\mathrm{st}(r_1, r_2)\,\gamma + \varsigma\,\mathbb{1}_{\mathrm{pair}}\,\gamma\right)\psi_i - \left(\delta + \delta_{\mathrm{rr}}\mathbb{1}_{\mathrm{rr}}\right)\mathrm{dom}(r_1, r_2)\,(1 - \mathbb{1}_{\mathrm{raiser}}) \\ - \lambda\,(12 - r_1)\,\mathbb{1}_{\mathrm{off}} - \beta\,(12 - r_1)\,\mathbb{1}_{\mathrm{pair}}, \tag{3.3}$$

with $\psi_i = \psi$ for an out-of-position caller and 1 otherwise, and finally

$$\rho_i = \max\left(\mathrm{pos}_i \cdot H_i \cdot (1 + \iota\,\mathbb{1}_{\mathrm{raiser}}),\ 0.05\right). \tag{3.4}$$

The floor keeps every factor positive, so the denominator of the share never vanishes; where it binds, ρ_i no longer depends on θ, so its gradient there is zero (Section 3.8).

Why this shape. The structure is chosen so that the solver produces *polarised* re-raising ranges, as real solvers do. Hands that realise badly when calling, such as dominated offsuit broadways, prefer to raise or fold. Hands whose value is postflop playability, such as suited aces, suited connectors and small pairs, get that value in deep pots, which mostly means calling. Suited wheel aces like A5s keep their straight and flush terms even as the aggressor, which makes them natural bluffs. An earlier model, with ρ a function of position and hand class only and shrunk towards 1 in shallow pots, barely depended on role and produced linear 3-betting ranges: small pairs looked like good 3-bets and A5s lost its edge. Figure 3.3 shows the resulting factors for four roles.

Neutral setting. With $\phi = \eta = \varsigma = \delta = \delta_{\mathrm{rr}} = \lambda = \beta = \iota = 0$, $\kappa = 0$ and $\psi = \xi = \chi = 1$, every $H_i = 1$ and $\rho_i \in \{1, \alpha_{\mathrm{oop}}\}$: a position-only model. We use it as a baseline in Section 4.3.

Symbol	Code name	Role and rationale
α_{oop}	`oop`	position factor out of position ($\alpha_{\text{ip}} = 1$ fixes the scale)
ϕ	`flush`	suited hands make flushes: a bonus at any depth
η	`conn`	per straight using both cards, scaled by γ: straights pay off only with money behind
ς	`set`	pairs make sets ('set mining'), scaled by γ
δ	`dom`	per point of domination, for callers only: a dominated caller meets the hands that crush it
δ_{rr}	`domPot`	extra domination penalty in re-raised pots, where the raiser's range is narrower
λ	`low`	offsuit non-pairs, per rank the top card sits below an ace
β	`pairLow`	pairs, per rank below aces: overcards come on most flops
κ	`spr`	half-life SPR of the depth fade γ
ι	`init`	bonus for the last aggressor (initiative)
ψ	`oopCall`	scales the playability bonuses of out-of-position callers
ξ	`callerShrink`	how much of the depth shrink of the position penalty an out-of-position caller gets
χ	`ipCall`	scales position for in-position callers in re-raised pots

Table 3.1: The realization parameters θ. Fitted values are in Appendix B.

3.4 The solver

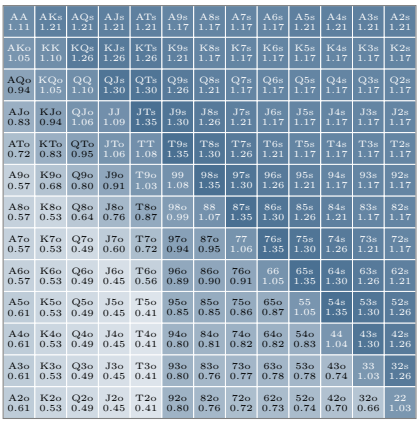
The solver implements external-sampling MCCFR (Algorithm 1) with linear averaging applied a block at a time (Proposition 2.10). The traverser cycles through the seats, the deal and every non-traverser action are sampled from their true probabilities, and the traverser enumerates its actions.

Data layout. Regrets and strategy sums are flat arrays of single-precision floats of shape [nodes][169][3], two tables of about 15 MB each. The tree is a few levels deep, so the traversal is a plain recursion over node indices, and each terminal's payoff inputs are read from the precomputed table of Section 3.1.

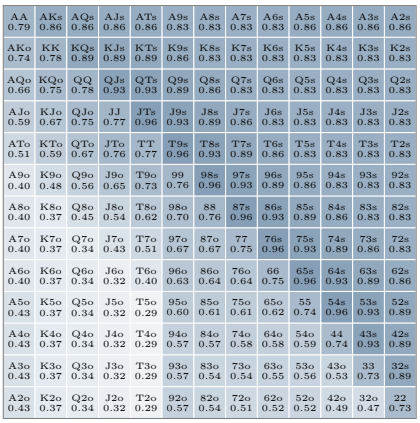
Lazy linear weighting. Linear CFR multiplies both tables by $k/(k+1)$ after every block of $N = 10{,}000$ iterations, which gives block j a weight proportional to j. Rather than sweep the tables, we add each increment of block j multiplied by j. The tables then hold the discounted sums times a common factor, and regret matching and the normalised average are both invariant to such a factor (Proposition 2.10), so the strategies are identical up to rounding and no pass over the tables is ever needed.

Threads. Iterations run in parallel in the style of Hogwild! [30]: all threads share the two tables and update them with relaxed atomic additions, without locks. Each thread has its own random number generator, seeded from the solver's. Because the lazy weights need no synchronised discount pass, threads never wait for each other. With one thread a run is bit-reproducible for a fixed seed; with more it is not, which is why our comparisons always use several seeds.

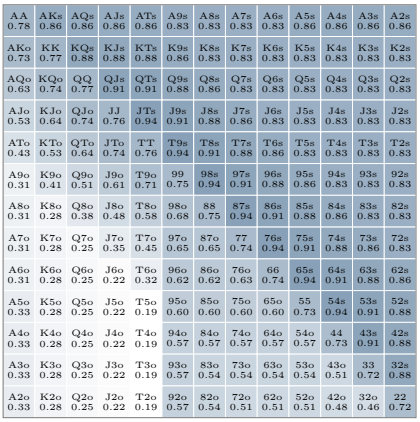
Average strategy. The output strategy is the strategy sum normalised per information set. Some information sets are reached only through their owner's own enumeration, after an action that player never takes, and accumulate no strategy sum; there we output the current regret-matched strategy, which is uniform only if the set was never reached. This matters for the trainer, whose bots must respond sensibly when the user deviates.



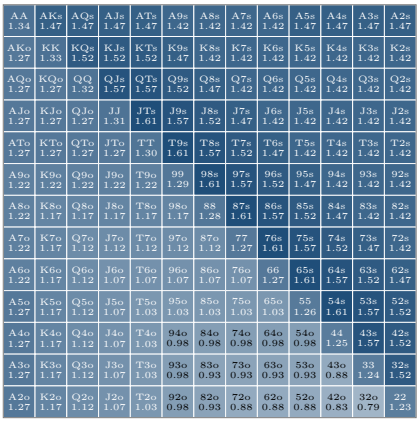
(a) In-position caller, single-raised pot (BTN calls a CO open), SPR = 15.0.



(b) Out-of-position caller, single-raised pot (BB calls a BTN open), SPR = 17.7.



(c) Out-of-position caller, 3-bet pot (CO calls a BTN 3-bet), SPR = 5.6.



(d) In-position raiser, 3-bet pot (the BTN 3-bettor in the same pot), SPR = 5.6.

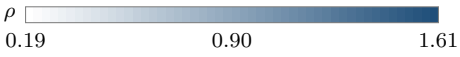


Figure 3.3: The realization factor ρ for every hand class under the adopted parameters, in four roles, on one shared colour scale. Each grid is the 13×13 class grid: pairs on the diagonal, suited hands above it, offsuit below.

Checkpoints. The full state (configuration, iteration count, random state and both tables) is written after every chunk of iterations, to a temporary file that is then renamed, so a crash never leaves a half-written file. A checkpoint made under different rules is refused. One made under different realization parameters is accepted as a *warm start*: the regrets are kept and the strategy sums and iteration count restart.

3.5 Two solves: Scratch and Anchored

Scratch is MCCFR from scratch over the whole tree with the fitted parameters, 3×10^8 iterations. It is the strategy we compare with the reference charts.

Anchored freezes the reference charts. At every (spot, class) pair a chart covers, the solver plays the chart's probabilities, both when sampling and when the actor is the traverser, and never updates the strategy sum, so the output keeps the chart exactly. Everything else, notably the squeezes, cold 4-bets and overcalls that the charts do not cover, is solved as usual and so becomes a best response to players who follow the charts. Anchored is warm-started from the Scratch checkpoint. Otherwise, after an action a chart never takes (a call in a 3-bet-or-fold chart,

say), the players behind reach that spot only when the locked player deviates, which happens only while the locked player is the traverser, so they would never accumulate regret there and would stay uniform.

The lock report. Regrets at locked entries are still accumulated, though never played. At the end of an Anchored solve, the mean regret of the best action against the chart's mix measures how much the chart gives up, in our model, against everything else. Summed over a spot's classes and ranked, these 'lock reports' point to where the model and the charts disagree most (Section 4.4).

3.6 Reference charts and the distance metric

Source. Our reference is the publicly available range viewer of PokerTrainer.se [29], which publishes GTO ('game-theory optimal', that is, approximate-equilibrium) charts, and a second PTO ('player-theory optimal') set adapted to typical opponents and so deliberately exploitative, for six-max 100 bb cash games with rake. Both sets cover the same 65 spots: 5 opens, 15 responses to an open, 15 responses to a 3-bet, 15 to a 4-bet and 15 to a 5-bet all-in. The charts are mixed: each cell gives a raise and a call frequency. We read them with a small browser-automation script and store them in a plain-text format with one file per spot. The bet sizes of our game are those of the charts (Appendix A), so differences are not sizing artefacts. All chart grids reproduced in this report are credited to this source.

Distance. We score a solved strategy $\bar{\sigma}$ against a chart at spot s with the weighted total-variation distance of Definition 2.13: each class k is weighted by its combo count n_k times the probability π_k that our strategy brings the class to the spot through the actor's own earlier actions, so classes we never bring there do not count. A spot's distance lies in $[0, 1]$, with 0 meaning identical play. Spots are grouped into the five categories above; the overall score averages the spots within each category, then averages the categories, so the fifteen deep 4-bet and 5-bet spots do not outweigh the five opens.

3.7 Tuning by coordinate search

The parameters θ are chosen to minimise the score plus a penalty for the UTG and BTN open frequencies leaving a band of ± 2.5 percentage points around the charts' own open widths. The search is a coordinate pattern search in the style of Hooke and Jeeves [16] (Section 2.7):

- each round evaluates $\theta \pm \delta_j e_j$ for every coordinate j, each candidate being a *fresh* full solve averaged over a few seeds, the same seeds for every candidate;
- every coordinate whose better step beats the current point by more than z standard errors moves, all together, falling back to the best single move if the combination is worse; the standard error comes from the run-to-run spread pooled over all candidates of the round;
- when nothing moves the steps halve and the iteration count may grow;
- at the end, the winner and the start point are re-scored at the final iteration count.

An earlier version evaluated candidates by short warm-started runs from a converged solve. That ranked candidates by how fast the solve moved away from the old solution rather than where it would end up, and it stalled; fresh solves are slower but measure the right thing. One round over thirteen parameters costs about 26 solves per seed, roughly ten minutes at 5×10^7 iterations. This cost, a full solve per candidate, is what motivates the cheaper method of the next section.

3.8 A fixed-strategy fit

Coordinate search (Section 3.7) pays a full solve for every candidate θ, which caps the number of realization parameters that can be tuned. The fixed-strategy fit trades that cost for an approximation: it holds every player's strategy fixed and asks only how the action values at the charted spots move with θ. The strategy held fixed is the Anchored solve (Section 3.5), which plays the charts exactly where they exist and the solved strategy elsewhere. Under it, the quantal-response loss of Section 2.7 becomes a cheap, smooth function of θ. Figure 3.4 shows the pipeline.

The sampler. For every charted spot s, every hand class k that the strategy brings to s, and every legal action a, the sampler estimates $\mathrm{EV}(s,k,a)$ like the trainer's rollouts (Section 2.6.3). It draws the hero's combination uniformly from class k and the opponents' hands from the remaining fifty cards, weights the deal by the likelihood ℓ of the opponents' actions on the way to s, and plays every action out on that deal, with everyone, the hero included, following the fixed strategy afterwards. Equities are computed once per deal and live set, and shared by the actions. Deals continue until the effective sample size reaches a target or a cap. A playout ends in one of three ways. Folds and pots won without a flop do not depend on θ, and nor do all-ins, which pay raw equity (Section 2.6.2). Only flop terminals do. The sampler therefore sums the first two kinds into a constant $F_{s,k,a}$, and keeps every flop terminal as a *record* r: its terminal z_r, the deal's weight ℓ_r, and each live player's class $k_{r,j}$ and equity $\mathrm{eq}_{r,j}$. With $W_{s,k} = \sum_n \ell_n$ over the deals and h the hero,

$$\mathrm{EV}_\theta(s,k,a) = \frac{1}{W_{s,k}}\Big(F_{s,k,a} + \sum_{r\in\mathcal{R}_{s,k,a}} \ell_r \,\mathrm{pay}_\theta(r)\Big), \tag{3.5}$$

$$\mathrm{pay}_\theta(r) = (\mathrm{pot}-\mathrm{rake})(z_r)\,\frac{\rho_h\,\mathrm{eq}_{r,h}}{\sum_j \rho_j\,\mathrm{eq}_{r,j}} - C_h(z_r),$$

the sum running over the live players at z_r. This is (2.6) with the flop terminals left open. Because the deals, boards and equities are fixed once sampled, (3.5) is a deterministic, almost everywhere smooth function of θ: every candidate is evaluated on the same sample, a sample-average approximation with common random numbers across θ.

The fitter. A realization factor depends on a record only through the player's class and their *role* at z_r: whether they act last after the flop, whether they made the last raise, whether the pot was re-raised, and the stack-to-pot ratio SPR. Far fewer roles than records occur, so ρ and its derivatives in θ are evaluated on a role-by-class table, which the records index into. The loss is $\mathcal{L}(\theta)$ of Section 2.7, with $\omega_{s,k}$ the class weights of Definition 2.13 balanced by category, plus $\nu\sum_m \big((\theta_m-\theta_{0,m})/\delta_m\big)^2$, where θ_0 is the sampling point and δ_m the coordinate-search step of parameter m. Its gradient is computed exactly in reverse order. The softmax gives $\partial\mathcal{L}/\partial\mathrm{EV}$, (3.5) gives $\partial\mathrm{EV}/\partial\,\mathrm{pay}_\theta(r) = \ell_r/W_{s,k}$, and with s_h the hero's share at z_r,

$$\frac{\partial\,\mathrm{pay}_\theta(r)}{\partial\rho_j} = (\mathrm{pot}-\mathrm{rake})(z_r)\,\frac{\mathrm{eq}_{r,j}\,(\mathbb{1}[j=h]-s_h)}{\sum_m \rho_m\,\mathrm{eq}_{r,m}}.$$

These are accumulated into the table and multiplied by the analytic $\partial\rho/\partial\theta$, which is zero where the floor of the realization model binds (Section 3.3). The optimiser is L-BFGS-B [8] with box bounds, in units of the steps δ_m. Records are sorted by (s,k,a) and split into chunks that threads process in parallel. A *hold-out* option leaves out the spots whose name matches a pattern and reports them separately, as a check that the fit generalises. The output is $\hat\theta$, the loss and the chart's regret by category before and after the fit.

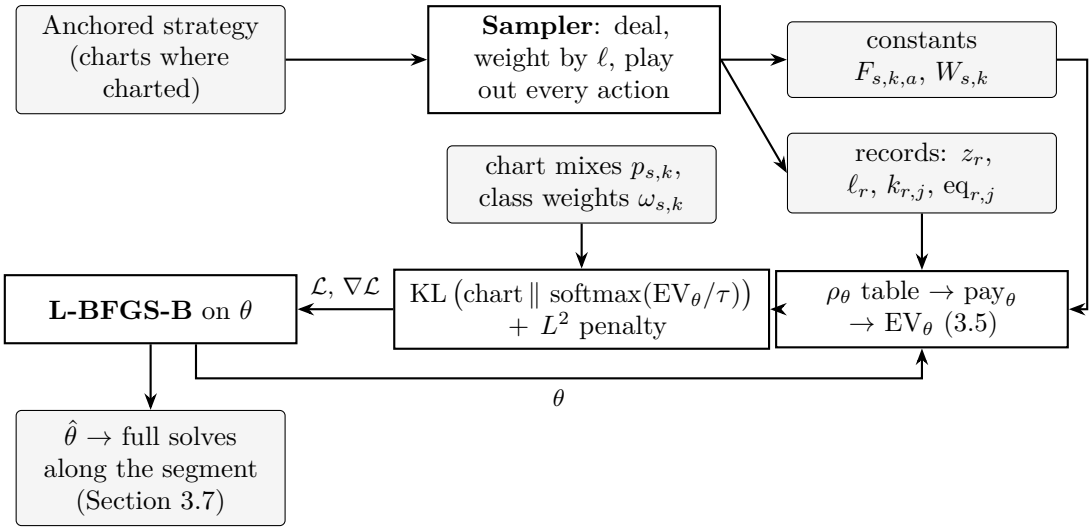


Figure 3.4: Data flow of the fixed-strategy fit. The sampler runs once, at θ_0; the loop below it re-evaluates the action values from the stored records for every θ the optimiser asks for, with no new playouts.

Checks. Three checks guard the pipeline. First, at the sampling point θ_0 the fitter recomputes every record's payoff and must reproduce the sampler's to floating-point precision, which validates the port of the realization model. Second, the sampler's estimates (3.5) at θ_0 must agree with the trainer's independent rollouts at a few spots within their standard errors (the first gate of the study). Third, a self-test writes a synthetic sample in the same format, sets each chart to $\mathrm{softmax}(\mathrm{EV}_{\theta^*}/\tau)$ for a planted θ^* away from the sampling point, and checks that the fit recovers θ^* to a small fraction of a step. It also checks the analytic gradient against finite differences.

What the fit cannot see. The fit asks whether the charts are quantal best responses to opponents who keep playing the fixed strategy. A full re-solve under $\hat{\theta}$ also moves the other players' ranges, most of all in 4-bet pots where the ranges are narrow, and the fit is blind to that. We therefore treat $\hat{\theta}$ as a direction, score points along the segment from θ_0 towards it with full solves and the distance of Definition 2.13, and adopt only what those confirm (Section 4.5).

3.9 The trainer

The trainer deals the user a spot, plays the five other seats as bots, and grades the user's decision. It has a headless core, which is unit-tested, and a graphical front end written with raylib [31].

Spots and bots. A spot of a requested type (open, facing an open, squeeze, facing a 3-bet, facing a squeeze, facing a 4-bet or more) and position is found by rejection sampling: deal a hand, let the bots and the user's own earlier decisions play from the strategy, and accept it when the user reaches a matching decision. Bots sample their actions from the strategy's average, so they stay mixed and respond sensibly to off-equilibrium play by the user.

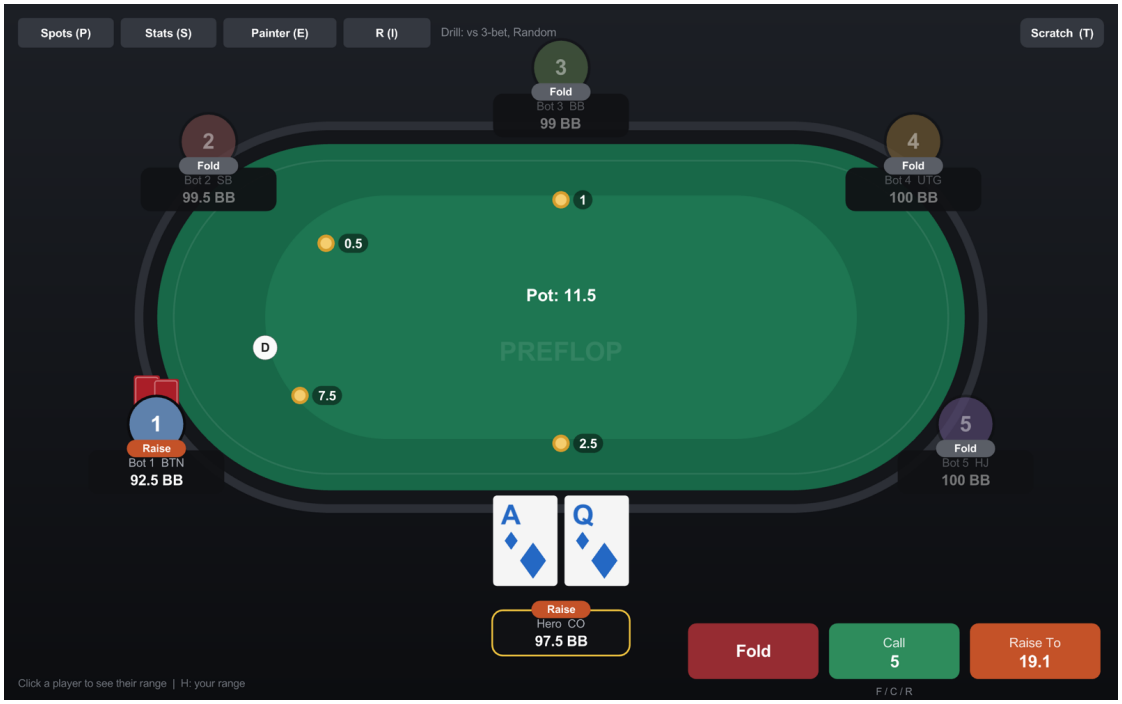
Rollout EVs. Reading action values from sampled regrets would be far too noisy. Instead, for the user's actual hand, the trainer re-deals the opponents' cards from the deck minus the user's, weights each deal by the probability that the bots would have produced the observed history with those cards, and plays every legal action out on the same deals with everyone following the strategy. This is the self-normalised importance-sampling estimator of Equation (2.6). Using the bots' actual cards would leak hidden information. The trainer keeps dealing until the effective sample size reaches 2000 (or a cap of two million deals), and it reports a standard error per

action and per difference between actions; since all actions share the deals, differences are far more precise than levels.

Grading. A choice is correct if its EV loss against the best action is at most 0.1 bb, or if the strategy plays it at least 10% of the time and loses at most 1 bb, or if the loss is within two standard errors. In the chart modes below, a decision covered by a chart is graded by the chart alone: correct if and only if the chart plays the action at least 10% of the time.

Modes. A key cycles three strategies: *Scratch*; *GTO*, which is Anchored with the GTO charts exactly where charted; and *PTO*, the PTO charts written over Anchored. Statistics (accuracy, average EV loss and the most-leaked spots) are kept per mode.

Interface. Figure 3.5 shows a hand from the trainer. The user, in the cutoff with A♢Q♢, opened and faces a 3-bet from the button; the buttons offer a fold, a call of 5 bb or a 4-bet to 19.1 bb. After the hand, the feedback panel shows the EV of each action with its standard error, the grade, the strategy's mix for the hand and the range for the whole spot. Other screens include a spot picker, statistics, a painter for editing charts and a realization inspector (Figure 3.6), which shows ρ for every class in a chosen role, pot type and SPR.



(a) The decision: CO holds AQs against the button's 3-bet to 7.5 bb.



(b) Feedback at the end of the hand. CO 4-bet (the strategy's most frequent action) and the button moved all-in; the panel shows CO's range facing the all-in, AQs's mix (fold 99.8%), and rollout EVs: calling loses 6.3 bb more than folding.

Figure 3.5: The trainer's table view and feedback panel, Scratch mode. Cell colours follow the action (raise, call, fold); classes the strategy never brings to the spot are greyed out.

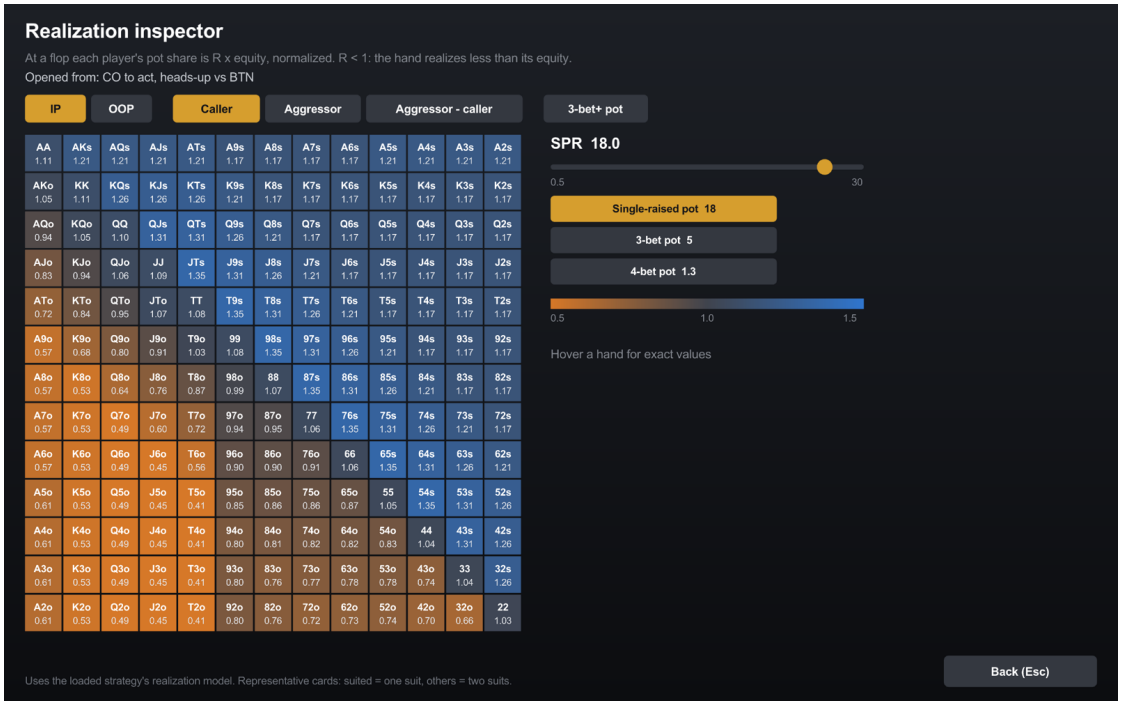


Figure 3.6: The realization inspector, showing ρ for an in-position caller in a single-raised pot at SPR 18. Suited connectors reach 1.35; dominated offsuit hands such as A9o fall to 0.57.

Chapter 4

Results and Discussion

All results use the shipped configuration: the adopted realization parameters (Appendix B), 3×10^8 iterations, $K = 100$ boards for multiway equity, and eight threads on a laptop (Apple silicon). Every number and figure in this chapter is regenerated from the code by one script.

4.1 Solver correctness on a solvable game

For six players CFR comes with no guarantee (Section 2.5), so we first check the solver where the theory applies. Forcing the first four seats to fold at the root turns our tree into a two-player game between SB and BB. With stacks of 10 bb and no rake it is a two-player zero-sum game in which SB moves all-in or folds and BB calls or folds: the classic *push/fold* game [12]. Its exploitability (Section 2.1.2) can be computed exactly from the heads-up equity table: for each player, the best response to the opponent's strategy is found class by class. We report the average of the two best-response gains, which is $\varepsilon/2$ in the sense of Definition 2.3; all exploitability figures in this section are on that scale.

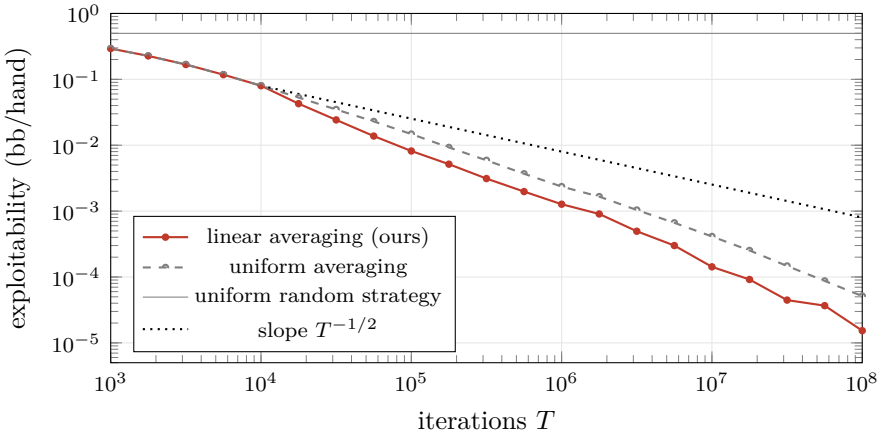


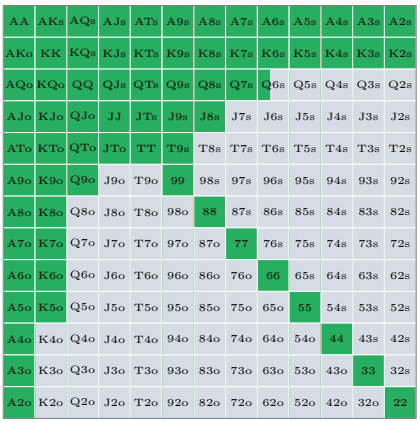
Figure 4.1: Exploitability of the average strategy in heads-up push/fold at 10 bb (no rake), reported as the mean best-response gain $\varepsilon/2$, mean of three seeds. The two schemes coincide for the first block of 10^4 iterations, which has weight 1 under both.

Figure 4.1 shows the exploitability of the average strategy against the number of iterations. Starting from 0.29 bb per hand after 10^3 iterations (a uniformly random strategy gives away 0.50 bb), it falls to 1.5×10^{-5} bb per hand after 10^8 iterations with linear averaging, a reduction by a factor of about 500 over the last three decades. This is close to a $1/T$ rate, much faster than the $O(1/\sqrt{T})$ worst-case bound of Section 2.2, as is commonly observed in practice. Linear averaging is about 3.4 times less exploitable than uniform averaging at 10^8 iterations, because it

discounts the poor early iterates. The resulting strategies (Figure 4.2) have the familiar shape: SB moves all-in with most hands, BB calls with a tighter range weighted towards pairs and aces.



(a) SB: all-in (red) or fold.



(b) BB facing the all-in: call (green) or fold.

Figure 4.2: The solved heads-up push/fold strategies at 10 bb after 10^8 iterations.

4.2 Convergence and performance of the six-max solve

For the six-player game we track the *strategy change* between successive chunks of 10^7 iterations: the reach- and range-weighted total-variation distance between the average strategies at the start and end of the chunk, so frequent spots and frequent hands count most. Without a convergence guarantee, the practical stopping rule is that this number is small and still falling. Figure 4.3 shows it falling steadily from 6.1% in the second chunk to 0.16% in the last, a factor of about 40, with no sign of cycling. Throughput is 5.5×10^6 iterations per second on average with eight threads (lower in the first chunk, while the tables are cold), so the full 3×10^8-iteration solve takes about a minute.

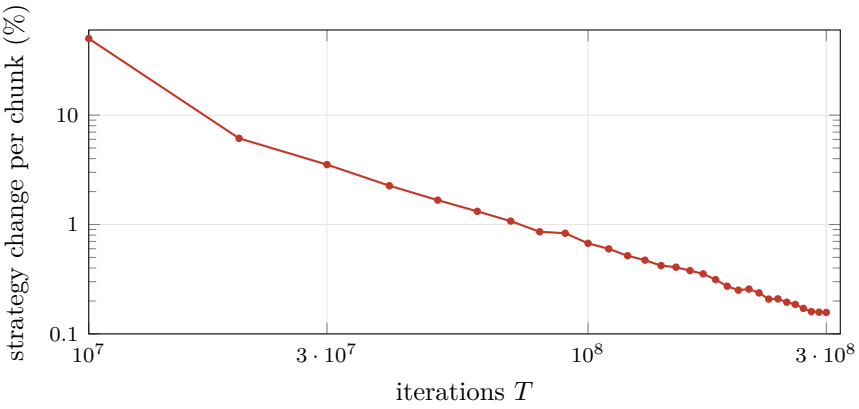


Figure 4.3: Strategy change of the six-max solve per chunk of 10^7 iterations. The first chunk is measured against the uniform initial strategy.

A second, model-internal check is indifference: at an equilibrium, every action a hand mixes between has the same value, and the strategy's mix loses nothing against the best action. The trainer's rollouts (Section 3.9) estimate exactly these action values for any dealt spot, with standard errors, so a user can spot-check the solution in any hand; we use this as a diagnostic rather than as a systematic metric.

4.3 Agreement with the reference charts

	n	Open	vs open	vs 3-bet	vs 4-bet	vs 5-bet	Overall
Against the GTO charts							
Position only (neutral)	1	0.119	0.166	0.334	0.569	0.406	0.319
Previous defaults	4	0.089±0.002	0.086±0.001	0.170±0.001	0.228±0.007	0.109±0.004	0.136±0.002
Adopted defaults	4	0.082±0.001	0.083±0.001	0.169±0.002	0.241±0.003	0.097±0.002	0.135±0.000
Adopted, shipped solve	1	0.085	0.083	0.166	0.244	0.101	0.136
Against the PTO charts							
Adopted, shipped solve	1	0.084	0.087	0.159	0.271	0.080	0.136

Table 4.1: Distance of the Scratch solve to the reference charts by category (lower is better; 0 is identical play), mean ± standard deviation over n seeds. 'Position only' sets every hand and role term of the realization model to zero (Section 3.3); 'previous defaults' are the coordinate-search values; 'adopted' are the shipped values (Appendix B).

Overall. Table 4.1 summarises the agreement. Against the GTO charts the shipped Scratch solve scores 0.136: 0.085 for opens, 0.083 for responses to an open, 0.166 facing a 3-bet, 0.244 facing a 4-bet and 0.101 facing a 5-bet all-in. Against the PTO charts it also scores 0.136. The realization model is what makes this possible. With a position-only model, the same solver scores 0.319, and 0.569 facing a 4-bet: without hand features, the solve cannot tell a hand that realises well from one that does not. The fitted features cut the overall distance by more than half, from 0.319 to 0.136; by category the cut ranges from about 30% for opens and 50% for responses to an open and to a 3-bet, to 57% facing a 4-bet and 75% facing a 5-bet. The open and flat/3-bet decisions, which are the most frequent in practice, agree best; the deep 4-bet spots, where every decision commits a large share of the stack, agree worst.

Sanity metrics. Table 4.2 lists the aggregate frequencies a player would check first, next to the same quantities computed from the GTO charts. Our open widths are within 1.5 percentage points of the charts' at every seat, from 17.8% under the gun (chart 16.3%) to 41.7% on the button (chart 42.1%). The big blind's total defence is close to the charts' as well, widening from 19% of hands against UTG to 35% against the button, but its composition is not: against the button the chart calls 26.4% and 3-bets 13.5%, while we call 15.9% and 3-bet 18.7%. Likewise, openers facing a big blind 3-bet fold about as often as the chart (58–65% of their range) but 4-bet more and call less, 20.0% against 12.0% on the button. Aces and kings always 4-bet. Our solve, in short, prefers raising to calling more than the charts do, a pattern the next section traces to the realization model.

		BB vs this open (%)			Opener vs BB 3-bet (%)		
Seat	Open %	Fold	Call	3-bet	Fold	Call	4-bet
UTG	17.8 (16.3)	80.9 (78.3)	12.1 (16.5)	7.0 (5.2)	63.9 (63.8)	16.8 (21.8)	19.3 (14.5)
HJ	22.0 (21.2)	77.7 (74.0)	13.7 (18.8)	8.7 (7.1)	63.1 (66.2)	17.7 (23.0)	19.2 (10.8)
CO	28.9 (27.7)	73.0 (69.4)	14.8 (22.5)	12.2 (8.1)	64.1 (65.4)	16.0 (19.5)	19.9 (15.1)
BTN	41.7 (42.1)	65.4 (60.1)	15.9 (26.4)	18.7 (13.5)	65.4 (66.5)	14.6 (21.5)	20.0 (12.0)
SB	43.3 (43.4)	43.1 (46.8)	33.6 (33.5)	23.3 (19.7)	58.0 (52.9)	15.3 (26.5)	26.8 (20.6)

Table 4.2: Sanity metrics of the shipped Scratch solve, with the GTO charts' values in grey: open frequency per seat, the big blind's response to that open, and the opener's response to a big blind 3-bet. Each is a percentage of the actor's range at the spot: every class weighted by its combinations and by the probability that the actor's own earlier actions bring it there (card removal ignored). Facing an open, the big blind has not acted, so these are percentages of all 1,326 combos.

Range charts. Figures 4.4 and 4.5 place our ranges next to the GTO charts for six representative spots; Appendix C has more, including the PTO charts. Cells are filled left to right in proportion to the raise, call and fold frequencies; white cells are hands the player never brings to the spot. The opens agree closely (Figure 4.4): the same hands open, and the differences are at the edges of the range. Facing an open, the cold-calling spot BTN versus CO agrees well, while BB versus BTN is among the weakest spots of its category. There our solve always 3-bets A9o–A7o and mostly 3-bets the middle pairs (99–66), which the chart mostly flats, and folds K9o, J9o and T9o, which the chart defends. Facing a 3-bet and a 4-bet (Figure 4.5) the overall structure matches, value hands raising and the strongest calling, but the frequencies of the marginal hands differ more.

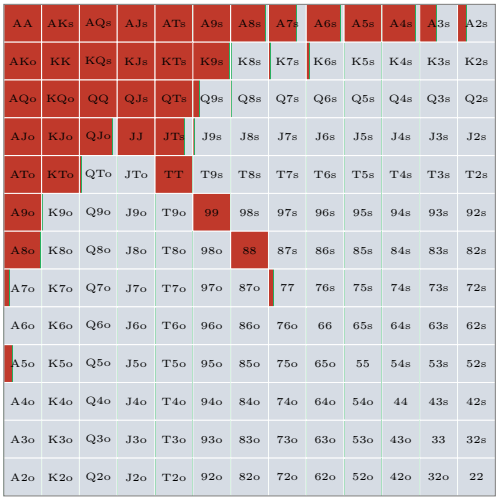
4.4 Where the model and the charts disagree

The Anchored solve (Section 3.5) plays the GTO charts exactly and solves everything else around them. Its lock report ranks the charted spots by how much the chart's play gives up, in our model, against the best response to everything else (Table 4.3). Two things stand out. First, the losses are small: the largest is 70 milli-big-blinds (mbb, thousandths of a big blind) per hand dealt, summed over all the classes of the spot, and no single class gives up more than about 2. Second, they are remarkably uniform in kind. In every one of the ten spots the better action is a *raise*, and the classes involved are offsuit aces and offsuit broadways: A9o–A6o defending the big blind against the button, AJo and AQo against a hijack open, and AJo, AQo, ATo and KQo 4-betting against a 3-bet.

Spot	Actor	Given up (mbb/hand)	Largest classes (better action, gain in mbb/hand)
BB vs BTN open	BB	70.16	A9o raise +2.12, A8o raise +2.05, A6o raise +1.71
BB vs HJ open	BB	54.40	AJo raise +1.17, AQo raise +1.16, A9o raise +1.15
SB vs BB 3-bet	SB	52.16	KTo raise +1.20, A9o raise +1.18, A5o raise +1.15
UTG vs CO 3-bet	UTG	48.50	AJo raise +1.54, AQo raise +1.46, KQo raise +1.12
UTG vs BTN 3-bet	UTG	45.08	AJo raise +1.70, AQo raise +1.52, ATo raise +1.44
HJ vs BTN 3-bet	HJ	33.22	AJo raise +1.63, KQo raise +1.48, ATo raise +1.42
HJ vs CO 3-bet	HJ	28.32	AJo raise +1.68, KQo raise +1.35, AQo raise +1.18
BB vs CO open	BB	26.26	ATo raise +0.85, A8o raise +0.84, A9o raise +0.84
CO vs BTN 3-bet	CO	23.74	AJo raise +1.68, KQo raise +1.56, ATo raise +1.40
SB vs BTN open	SB	22.67	A9o raise +0.89, A8o raise +0.81, A7o raise +0.79

Table 4.3: The ten charted spots where the GTO chart gives up the most against the rest of the Anchored solve, in our model. The loss is the best action's mean regret against the chart's mix, in thousandths of a big blind per hand dealt, summed over the spot's classes, so frequent spots and classes weigh more.

This is a systematic disagreement, and the structure of the realization model suggests where it comes from. The low-card term $\lambda(12 - r_1)$ is zero for any hand containing an ace, so weak offsuit aces are penalised only through domination, and the domination term applies only to *callers*. As the raiser, the same hand loses the domination penalty and gains the initiative bonus ι, while its ace blocks the opponents' strongest hands exactly, since the solver deals real cards. The model therefore makes raising, rather than calling or folding, the natural use of these hands, more so than the charts do, and the effect is largest in the deep 3-bet and 4-bet pots. A kicker-aware term that also applies to the raiser, or a domination penalty that depends on the opponent's range rather than only on the role, are the natural fixes. The largest differences in Figure 4.4 are consistent with the same pattern.



Scratch: UTG open $D = 0.053$



GTO chart [29]: UTG open



Scratch: BTN open $D = 0.094$



GTO chart [29]: BTN open



Scratch: BB vs BTN open $D = 0.185$



GTO chart [29]: BB vs BTN open

Figure 4.4: Scratch (left) against the GTO chart (right): opens and the big blind's defence. D is the spot's distance (Section 3.6). Raise in red, call in green, fold in grey; white: the hand never reaches the spot. Chart data from PokerTrainer.se [29].



Scratch: BTN vs CO open $D = 0.060$



GTO chart [29]: BTN vs CO open



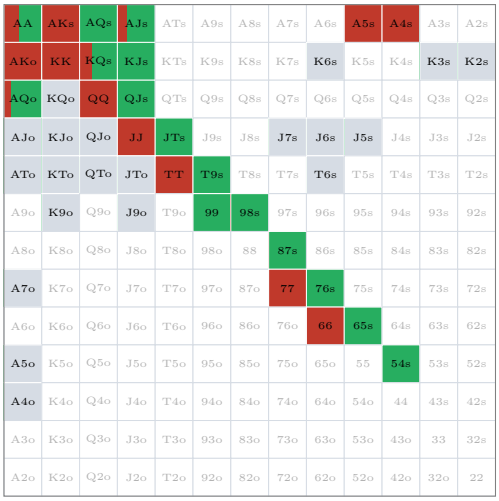
Scratch: CO vs BTN 3-bet $D = 0.175$



GTO chart [29]: CO vs BTN 3-bet



Scratch: BB vs BTN 4-bet $D = 0.275$



GTO chart [29]: BB vs BTN 4-bet

Figure 4.5: Scratch (left) against the GTO chart (right): a cold-call spot, facing a 3-bet and facing a 4-bet. Conventions as in Figure 4.4.

4.5 The fixed-strategy fit

This section reports the study of Section 3.8: fitting θ without a solve per candidate, by holding every strategy fixed at the Anchored solve and asking which θ makes the charts' own mixes look like (quantal) best responses. The study was run with the parameters found by coordinate search (Appendix B) and the Anchored solve made with them.

Gate 1: the sampler. The sampled records must reproduce the trainer's rollouts, which are computed independently. Table 4.4 compares the expected values rebuilt from the records with `rollout()` for 13 actions at 8 spot–class pairs, chosen to include multiway flops and all-ins, here regenerated on the current Anchored solve. The largest discrepancy is 1.8 combined standard errors (2.2 in the study's own run), consistent with sampling noise; folds agree exactly and are omitted. The working sample, at an effective sample size of 5000 per spot–class pair, holds 2.8×10^7 flop records over 1,029 flop terminals and 5,058 spot–class pairs. The numpy port of the payoff reproduces the C++ payoffs to 7×10^{-8} of the pot, and one evaluation of the loss and its analytic gradient takes about half a second.

Spot	Class	Action	EV from records (bb)	Rollout EV (bb)	z
UTG open	76s	raise	-0.296 ± 0.030	-0.284 ± 0.028	-0.30
BTN open	A5o	raise	0.297 ± 0.012	0.286 ± 0.012	$+0.65$
BTN vs CO open	T9s	call	-0.091 ± 0.053	-0.126 ± 0.031	$+0.58$
BTN vs CO open	T9s	raise	0.154 ± 0.042	0.242 ± 0.041	-1.49
CO vs UTG open	88	call	0.252 ± 0.080	0.053 ± 0.078	$+1.78$
CO vs UTG open	88	raise	-0.071 ± 0.044	0.035 ± 0.057	-1.47
BB vs BTN open	KJo	call	-0.617 ± 0.005	-0.620 ± 0.005	$+0.37$
BB vs BTN open	KJo	raise	0.331 ± 0.036	0.328 ± 0.036	$+0.06$
CO vs BTN 3-bet	AJs	call	-0.925 ± 0.016	-0.939 ± 0.017	$+0.61$
CO vs BTN 3-bet	AJs	raise	1.236 ± 0.072	1.137 ± 0.073	$+0.97$
BB vs BTN 4-bet	AKo	call	-4.968 ± 0.053	-5.045 ± 0.053	$+1.02$
BB vs BTN 4-bet	AKo	raise	0.355 ± 0.201	0.085 ± 0.201	$+0.95$
CO vs BB 5-bet	QQ	call	-2.739 ± 0.314	-3.158 ± 0.315	$+0.94$

Table 4.4: Gate 1: expected values rebuilt from the sampler's records against the trainer's independent rollouts, both at an effective sample size of 20,000, with standard errors; z is the difference over the combined standard error.

The fit. At temperature $\tau = 1$ bb, L-BFGS-B converges in 42 iterations (21 seconds). The balanced KL divergence falls from 0.770 to 0.693 and the charts' regret from 0.462 to 0.383 bb (Table 4.5). The gains are concentrated in the responses to a 3-bet (regret $0.89 \to 0.54$ bb) and to an open ($0.14 \to 0.08$ bb). Facing a 4-bet nothing improves (regret $0.91 \to 0.93$ bb, slightly worse), and facing a 5-bet nothing can, since those spots end in all-ins whose payoffs do not involve ρ. The fitted parameters move consistently across temperatures (Table 4.6): the flush, set, domination, low-card and re-raised-domination terms all grow. The exception is the depth half-life κ, which runs to its lower bound for $\tau \leq 1$, switching the depth fade off, but stays near its old value at $\tau = 3$.

Gate 2: full solves. The fit is only useful if it improves the real objective, the distance of a fresh Scratch solve to the charts. Table 4.7 reports full 3×10^8-iteration solves for each candidate. The fitted parameters make the solve *worse*: 0.149 and 0.151 at $\tau = 1$, and 0.148 and 0.145 at $\tau = 0.3$, against a mean of 0.136 for the starting point. The per-category breakdown shows why. The fit improves the opens and the responses to an open, as it promised, but the responses to a 4-bet deteriorate from about 0.23 to 0.29, which more than cancels the gains.

		KL divergence		Chart regret (bb)	
Category	Spots	before	after	before	after
open	5	0.5994	0.5927	0.0219	0.0195
vs_open	15	0.8067	0.7406	0.1415	0.0826
vs_3bet	15	1.0357	0.7743	0.8879	0.5408
vs_4bet	15	1.0895	1.0371	0.9130	0.9267
vs_5bet	15	0.3197	0.3197	0.3443	0.3443
Overall (balanced)	65	0.7702	0.6929	0.4617	0.3828

Table 4.5: The fixed-strategy fit at $\tau = 1$: the loss (KL divergence of the chart mixes from the logit response) and the charts' regret, by category, at the starting point and at the fitted $\hat{\theta}$, with all strategies held at the Anchored solve.

Parameter	Start	$\tau = 0.3$	$\tau = 1$	$\tau = 3$
α_{oop} (oop)	0.7	0.707	0.712	0.6829
ϕ (flush)	0.16	0.1826	0.1737	0.1664
η (conn)	0.05	0.05769	0.05082	0.04969
ς (set)	0.1	0.2373	0.1462	0.1092
δ (dom)	0.1	0.1049	0.1174	0.1176
λ (low)	0.03	0.06336	0.04895	0.03825
κ (spr)	3	0.05	0.05	2.668
ι (init)	0.2	0.2049	0.2355	0.2201
β (pairLow)	0.01	0.0129	0.004248	0.004839
ψ (oopCall)	1	1.13	1.045	1.006
ξ (callerShrink)	0	0	0	0
χ (ipCall)	0.8	0.8378	0.869	1.05
δ_{rr} (domPot)	0	0.07601	0.06649	0.07895

Table 4.6: Fitted realization parameters at three temperatures ($\alpha_{\text{ip}} = 1$ fixed). The depth half-life κ runs to its lower bound 0.05 for $\tau \leq 1$.

Partial moves. If the fit points in a useful direction but overshoots, a smaller step may help. Moving a quarter of the way changes nothing measurable. Moving halfway gives 0.1346 (mean of four seeds) against 0.1363 for the starting point, an improvement of 0.0017, about two standard errors, with the smallest seed-to-seed spread of all candidates. Moving all the way except for κ gives 0.1350, with a larger spread. Since the halfway point is the best of five candidates, part of its margin may be selection, and the gain should be read as small rather than established. The category means show a common pattern: moving towards the fit helps opens and responses to an open and hurts responses to a 4-bet; the halfway point also helps responses to a 5-bet. We adopted the halfway point, the values now shipped; the shipped solve, a single seed, scores 0.1357.

Why the fit fails. We conjecture the following explanation. The fixed-strategy loss asks only whether each chart is a good response to opponents who also play the charts. In a full solve every player adapts to the new θ at once. Changing the domination and depth terms changes which hands 4-bet, and the players facing a 4-bet then respond to a different range than the chart assumes, so their solved ranges drift away from the charts. The loss is blind to this equilibrium feedback, and the feedback is strongest exactly in the deep spots. A second symptom points the same way: at the fitted optimum the thirteen parameters do not make the 4-bet charts rational even holding everything else fixed (their KL barely moves), so small local gains elsewhere may not survive re-equilibration. The fixed-strategy fit is therefore a useful search direction, cheap enough to explore, but not a substitute for scoring full solves.

Candidate	n	Open	vs open	vs 3-bet	vs 4-bet	vs 5-bet	Overall
Previous defaults	4	0.089±0.002	0.086±0.001	0.170±0.001	0.228±0.007	0.109±0.004	0.136±0.002
Fit, $\tau = 1$	2	0.075±0.001	0.083±0.001	0.181±0.001	0.295±0.008	0.115±0.001	0.150±0.002
Fit, $\tau = 0.3$	2	0.070±0.002	0.079±0.000	0.178±0.003	0.291±0.002	0.113±0.008	0.146±0.002
25% towards fit	2	0.086±0.000	0.084±0.001	0.170±0.002	0.234±0.002	0.109±0.002	0.137±0.001
50% towards fit (adopted)	4	0.082±0.001	0.083±0.001	0.169±0.002	0.241±0.003	0.097±0.002	0.135±0.000
Fit with κ kept at 3	4	0.082±0.001	0.083±0.001	0.171±0.002	0.231±0.006	0.108±0.003	0.135±0.002

Table 4.7: Gate 2: distance to the GTO charts of fresh 3×10^8-iteration Scratch solves for each candidate, mean ± standard deviation over n seeds. The partial moves interpolate linearly between the previous defaults and the fit at $\tau = 1$.

4.6 Limitations

- **A model, not the game.** We solve a preflop game whose postflop value is replaced by thirteen parameters. However well they are fitted, the solution is an equilibrium of the model; agreement with the charts is evidence of a good model, not a proof of equilibrium play in real poker.
- **No guarantee beyond two players.** For six players and with rake, CFR need not converge to a Nash equilibrium (Section 2.5). We monitor convergence empirically and verify the solver on a two-player game, but cannot certify the six-player solution.
- **Equity simplifications.** The exact heads-up table ignores the cards of folded players, whose hands skew weak; we have not quantified the resulting dead-card effect. Multiway equity uses $K = 100$ sampled boards, so the realization-adjusted share carries an $O(1/K)$ bias (Section 2.6.2).
- **Abstraction.** One raise size per level, no limping and a single stack depth. These match the reference charts but restrict what the model can express.
- **The reference.** The charts are themselves the output of a solver with its own abstractions; a distance of zero is not the goal in itself, and the PTO charts are deliberately not an equilibrium.
- **Fitting cost.** Each candidate in the coordinate search is a full solve, which limits how many parameters can be fitted, and the cheaper fixed-strategy fit cannot see equilibrium feedback (Section 4.5).

4.7 Conclusion and future work

We built a complete pipeline for six-max preflop strategy: a precise preflop game with rake, a realization-adjusted payoff model built from interpretable hand features, an external-sampling Monte Carlo CFR solver that solves the 7,499-node tree in about a minute on a laptop, a fit of the model to published charts, and an interactive trainer that grades decisions with importance-sampled rollouts. The solver converges as theory predicts on the heads-up push/fold game, its six-player solve stabilises steadily, and the solution agrees with the published GTO charts to a category-balanced distance of 0.136, best for opens and responses to an open and worst facing 4-bets. A solve with the charts locked shows where the model and the charts disagree, chiefly in how often weak offsuit aces and broadways should re-raise. A cheaper fixed-strategy fit improved the frequent spots but could not see the equilibrium feedback that governs the deep ones; moving halfway towards it gave a small improvement of about two standard errors.

Several directions follow naturally.

- **A loss that sees re-equilibration.** The fixed-strategy loss could re-solve the responders'

ranges for each candidate, for instance the 4-bettor's range, which would make it track the full-solve objective in the deep spots. That is the precondition for fitting a richer model.

- **Richer realization features.** We have computed per-class features against a random hand (equity, the spread of flop equity, and made-hand and nut-flush frequencies) and the live-player count per terminal is available; a log-linear model $\log \rho = w^\top \varphi$ over them is the natural next model once the loss above exists.
- **Postflop ground truth.** Realization factors could be measured directly with a postflop solver on sampled flops and regressed on hand features, using the charts only for validation.
- **Game extensions.** Other stack depths need only a change of configuration; dead-card-aware equity, a second raise size and SB limping need code changes. All would bring the game closer to real play.

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Appendix A

Bet sizes and game parameters

All amounts are in big blinds. Raise totals round to the nearest 0.1 bb (halves away from zero) and are capped at the stack. The sizes are those of the reference charts [29]. A single out-of-position 4-bet multiplier cannot reproduce both of the charts' sizes of 19 and 23 after rounding; 2.55 gives 19.1 and 23.

Action	Rule	Examples
Open	2.5; 3 from SB	UTG opens to 2.5; SB opens to 3
3-bet in position	3× the open	BTN vs a CO open: 7.5; BB vs an SB open: 9
3-bet from the blinds	5× the open	BB vs a BTN open: 12.5
Squeeze	as a 3-bet, +1 open per caller	BTN vs HJ open, CO call: 10; BB vs CO open, two calls: 17.5
4-bet in position	2× the 3-bet	BTN vs BB's 12.5: 25
4-bet out of position	2.55× the 3-bet	CO vs BTN's 7.5: 19.1; SB vs BB's 9: 23
5-bet	all-in	100
Stacks, blinds	100; 0.5/1	reset every hand
Rake	5% of the pot, capped at 4	only if a flop is dealt or an all-in is called
Limping	not allowed	
Cold calls	of an open only	a cold caller may also call a squeeze

Table A.1: Bet sizes and rules of the game.

Appendix B

Fitted realization parameters

Table B.1 lists the realization parameters (Table 3.1) at three points: the values found by the coordinate search of Section 3.7, the optimum of the fixed-strategy fit at temperature $\tau = 1$ (Section 3.8), and the adopted values, halfway between the two (Section 4.5). The scale is fixed by $\alpha_{\text{ip}} = 1$.

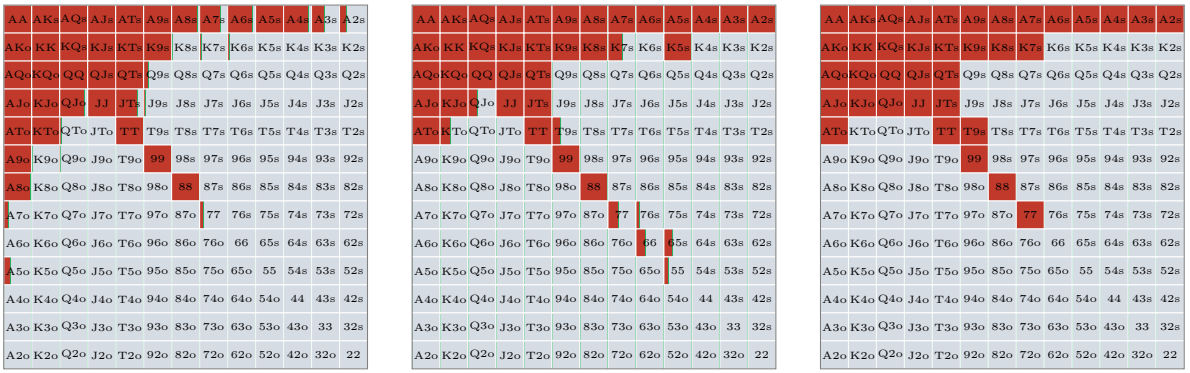
Symbol	Code name	Coordinate search	Fixed-strategy fit	Adopted
α_{oop}	`oop`	0.7	0.712	0.706
ϕ	`flush`	0.16	$0.173,7$	$0.166,9$
η	`conn`	0.05	$0.050,8$	$0.050,4$
ς	`set`	0.1	$0.146,2$	$0.123,1$
δ	`dom`	0.1	$0.117,4$	$0.108,7$
λ	`low`	0.03	$0.049,0$	$0.039,5$
κ	`spr`	3	0.05	1.525
ι	`init`	0.2	$0.235,5$	$0.217,8$
β	`pairLow`	0.01	$0.004,2$	$0.007,1$
ψ	`oopCall`	1	1.045	$1.022,5$
ξ	`callerShrink`	0	0	0
χ	`ipCall`	0.8	0.869	$0.834,5$
δ_{rr}	`domPot`	0	$0.066,5$	$0.033,2$

Table B.1: Realization parameters. The fixed-strategy fit drove κ to its lower bound 0.05, which switches the depth fade off.

Appendix C

Further range charts

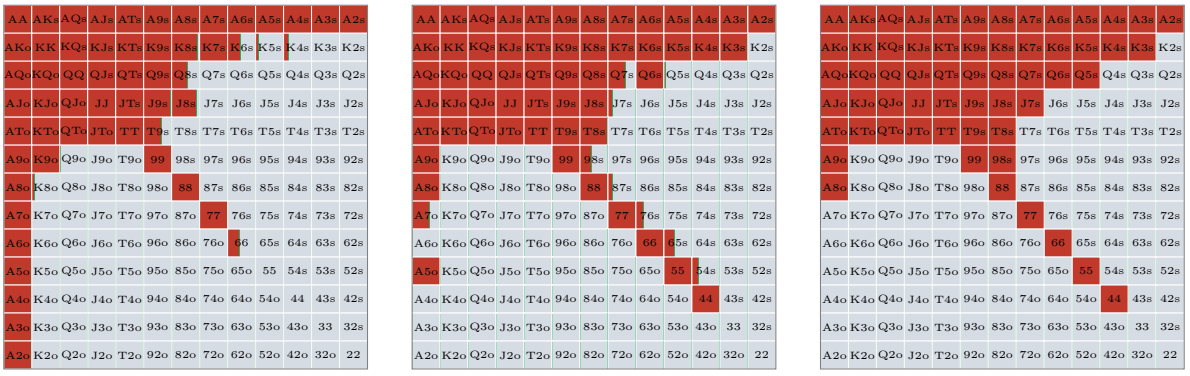
This appendix shows, for fourteen spots, the Scratch solve next to the GTO and PTO charts of PokerTrainer.se [29], with the distance D of the Scratch solve to each chart (Section 3.6). Conventions are as in Figure 4.4: raise in red, call in green, fold in grey, and white for hands the player never brings to the spot. In the chart grids the reach is the chart's own.



Scratch: UTG open GTO $D = 0.053$ PTO $D = 0.052$



Scratch: HJ open GTO $D = 0.066$ PTO $D = 0.059$



Scratch: CO open GTO $D = 0.088$ PTO $D = 0.108$



Scratch: BTN open GTO $D = 0.094$ PTO $D = 0.094$

Figure C.1: Scratch, GTO and PTO: UTG open, HJ open, CO open, BTN open. Chart data from PokerTrainer.se [29].



Scratch: SB open GTO $D = 0.122$ PTO $D = 0.105$



Scratch: CO vs UTG open GTO $D = 0.052$ PTO $D = 0.050$

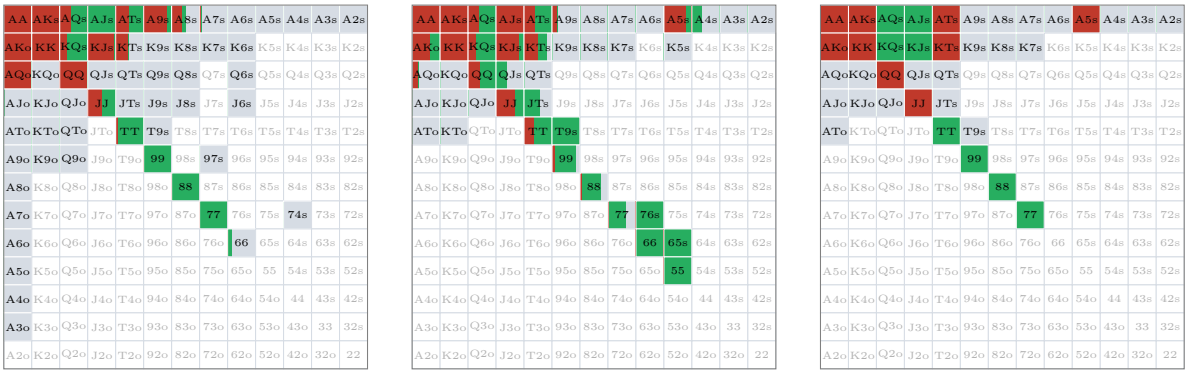


Scratch: SB vs BTN open GTO $D = 0.045$ PTO $D = 0.045$



Scratch: BB vs SB open GTO $D = 0.202$ PTO $D = 0.199$

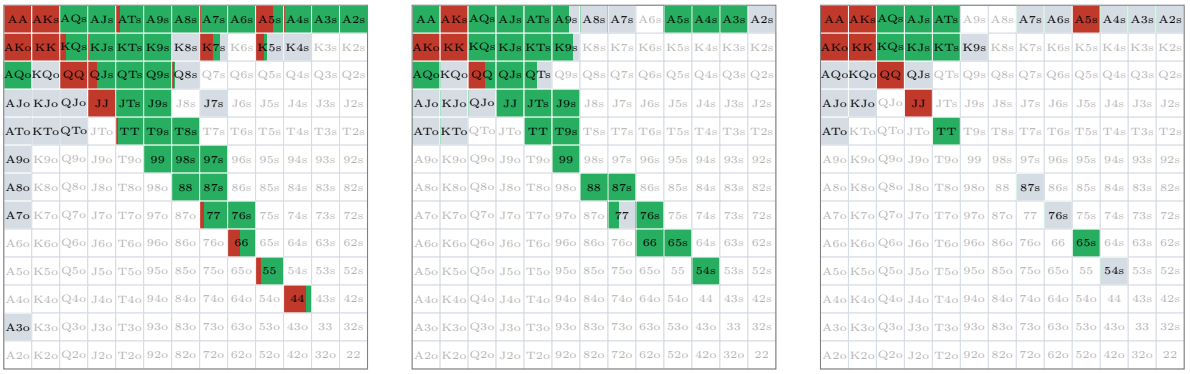
Figure C.2: Scratch, GTO and PTO: SB open, CO vs UTG open, SB vs BTN open, BB vs SB open. Chart data from PokerTrainer.se [29].



Scratch: UTG vs BTN 3-bet GTO $D = 0.185$ PTO $D = 0.157$



Scratch: SB vs BB 3-bet GTO $D = 0.197$ PTO $D = 0.144$



Scratch: BTN vs CO 4-bet GTO $D = 0.229$ PTO $D = 0.409$



Scratch: SB vs BTN 4-bet GTO $D = 0.206$ PTO $D = 0.264$

Figure C.3: Scratch, GTO and PTO: UTG vs BTN 3-bet, SB vs BB 3-bet, BTN vs CO 4-bet, SB vs BTN 4-bet. Chart data from PokerTrainer.se [29].

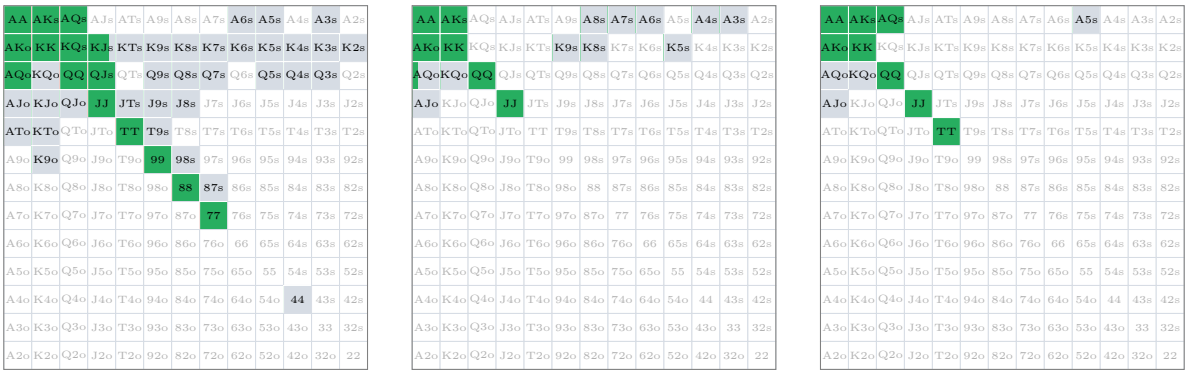


Figure C.4: Scratch, GTO and PTO: BTN vs BB 5-bet, CO vs SB 5-bet. Chart data from PokerTrainer.se [29].